

MSP Report - Sprint 2

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Github (Sprint 1): https://github.com/niklassander/MSP/releases/tag/Sprint1_Submission

Github (Sprint 2): https://github.com/niklassander/MSP/releases/tag/Sprint2_Submission

Roles

Scrum Master: Franka Traupe

Product Owner: Guilherme Fernandes

Quality Controllers and Development Team: Franka Traupe, Guilherme Fernandes, Roschanak Babai, Valeriia Baida, Niklas Sander

Features

(Implemented features highlighted in bold)

Basic Features (Implemented in prototype)

- **Register account**
- **Register type of vehicle**
- **Define route**
- **Alert if speed limit is violated**
- **Calculate estimated time of arrival**

Features from Existing System

- **Bookmarking (home, work, custom)**
- **Avoid tolls**
- **Recent destinations**
- **Show current speed (implemented already as part of speed alerts)**
- **Parking lots**
- **Hazard reports**
- **Change units**

New Features

- **Statistics: usage and driving quality (distance traveled, active days etc.)**
- **Walking route**
- **Shared ride**

Project Planning: Sprint 1

Task Table

Task ID	Task	Duration (days)	Depends on
1	Explore original app & choose features	3	-
2	Review chosen features	1	1
3	Develop BPMNs for each feature	4	2
4	Review BPMNs	1	3
5	Front-end development	7	4
6	Back-end development	7	4
7	Integrate front-end and back-end	2	5,6
8	Testing	2	7
9	Bug Fixing	1	8

Table 1: Task List with Dependencies

Gantt Chart

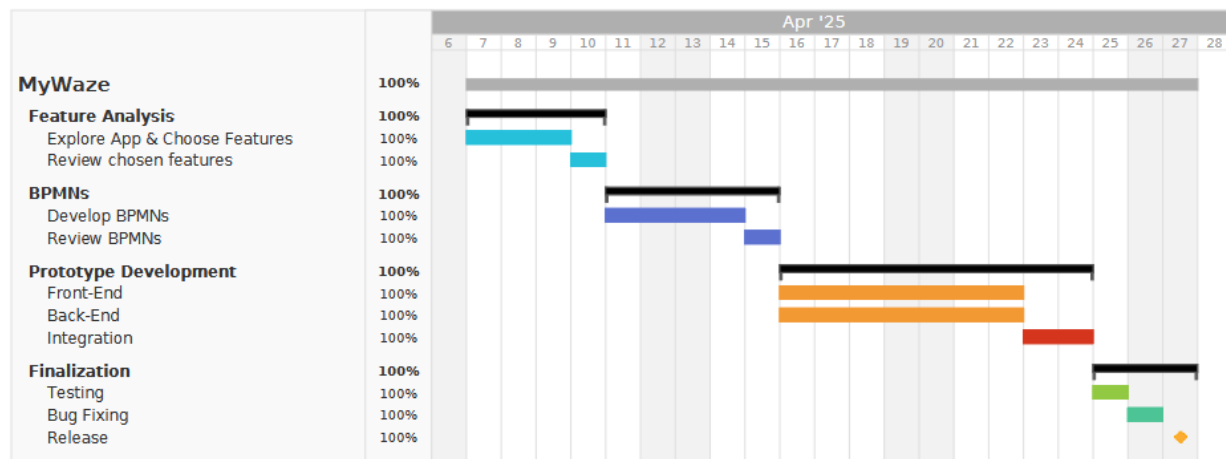


Figure 1: Gantt Chart

Burndown Chart

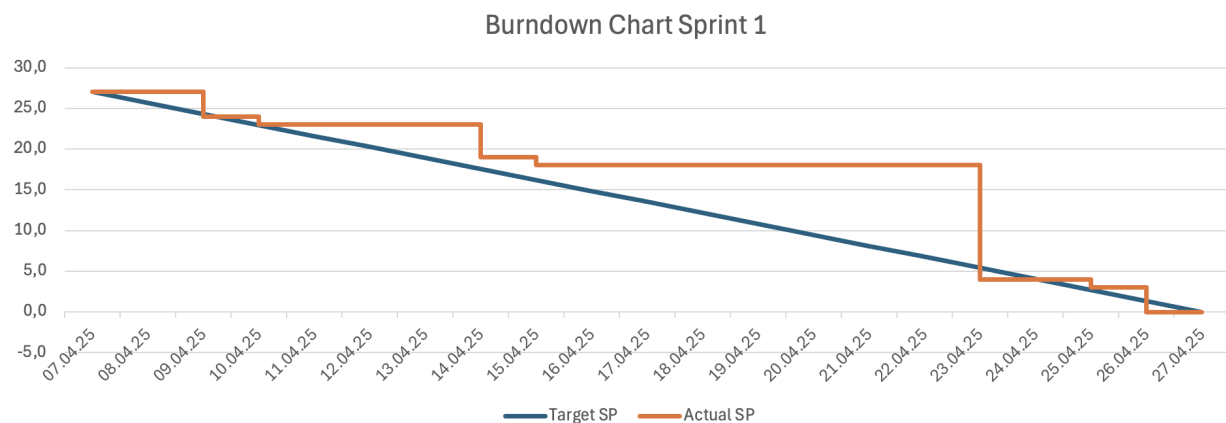


Figure 2: Burndown Chart

Project Planning: Sprint 2

Task Table

Task ID	Task	Duration (days)	Depends on
1	Sprint 1 Revision	2	-
2	Context Diagram	3	1
3	Use Case View	3	1
4	Logical View	3	1
5	Development View	4	4
6	Process View	11	5
7	Prototyping	11	5
8	Deployment View	3	7

Table 2: Task List with Dependencies

Gantt Chart

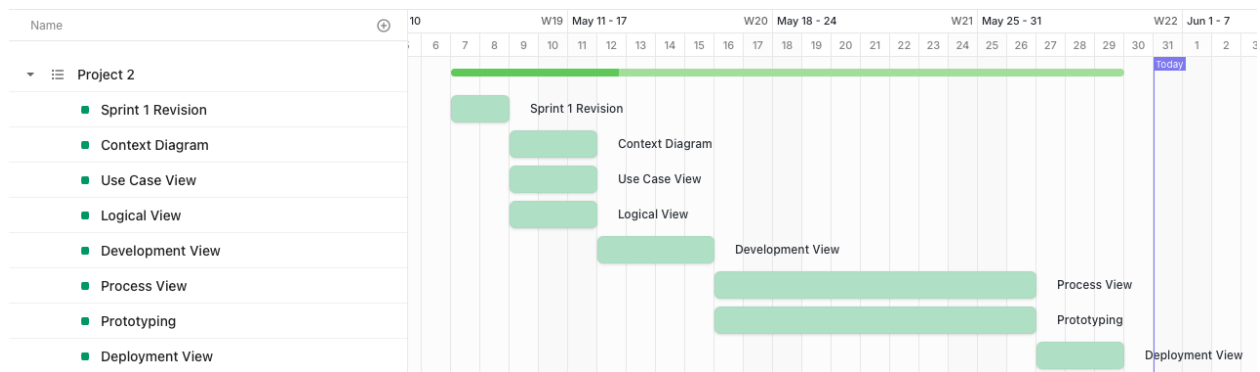


Figure 3: Gantt Chart

Burndown Chart

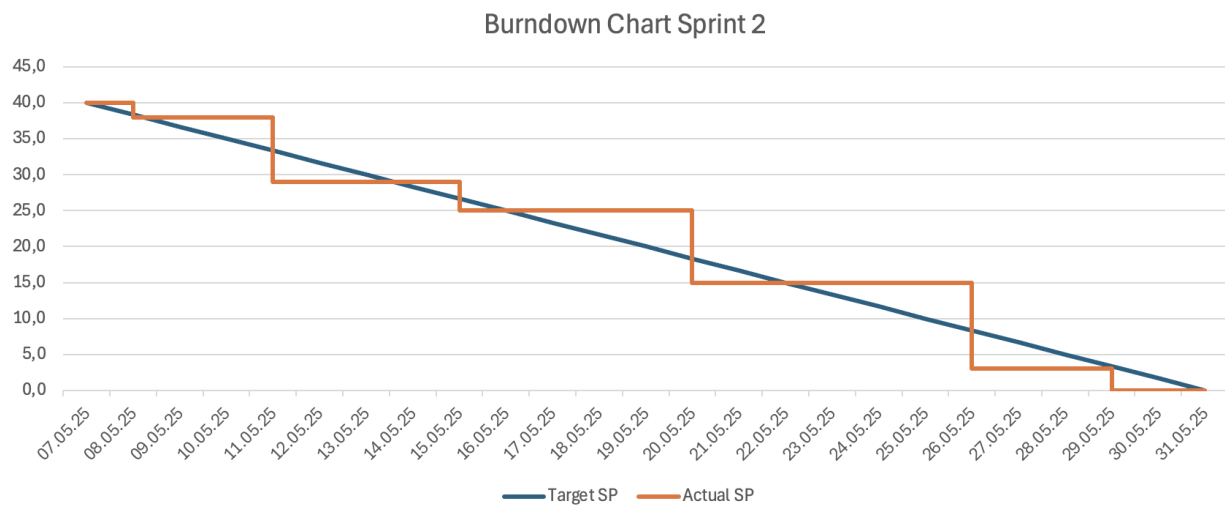


Figure 4: Burndown Chart

Requirements Diagrams

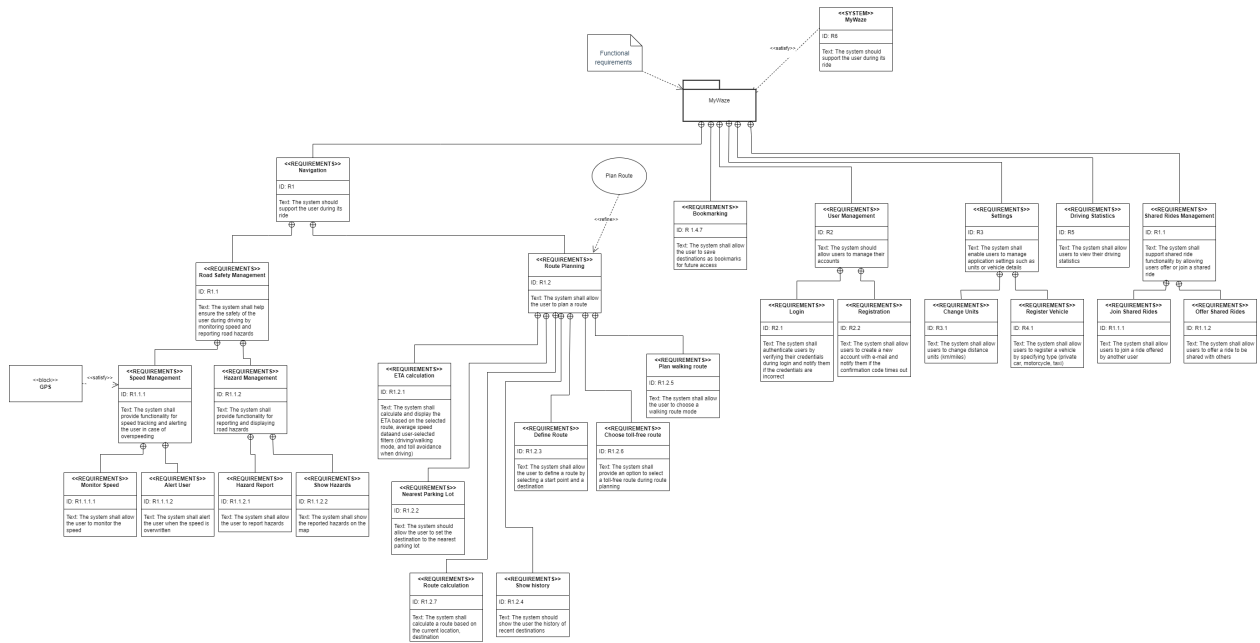


Figure 5: Functional Requirements

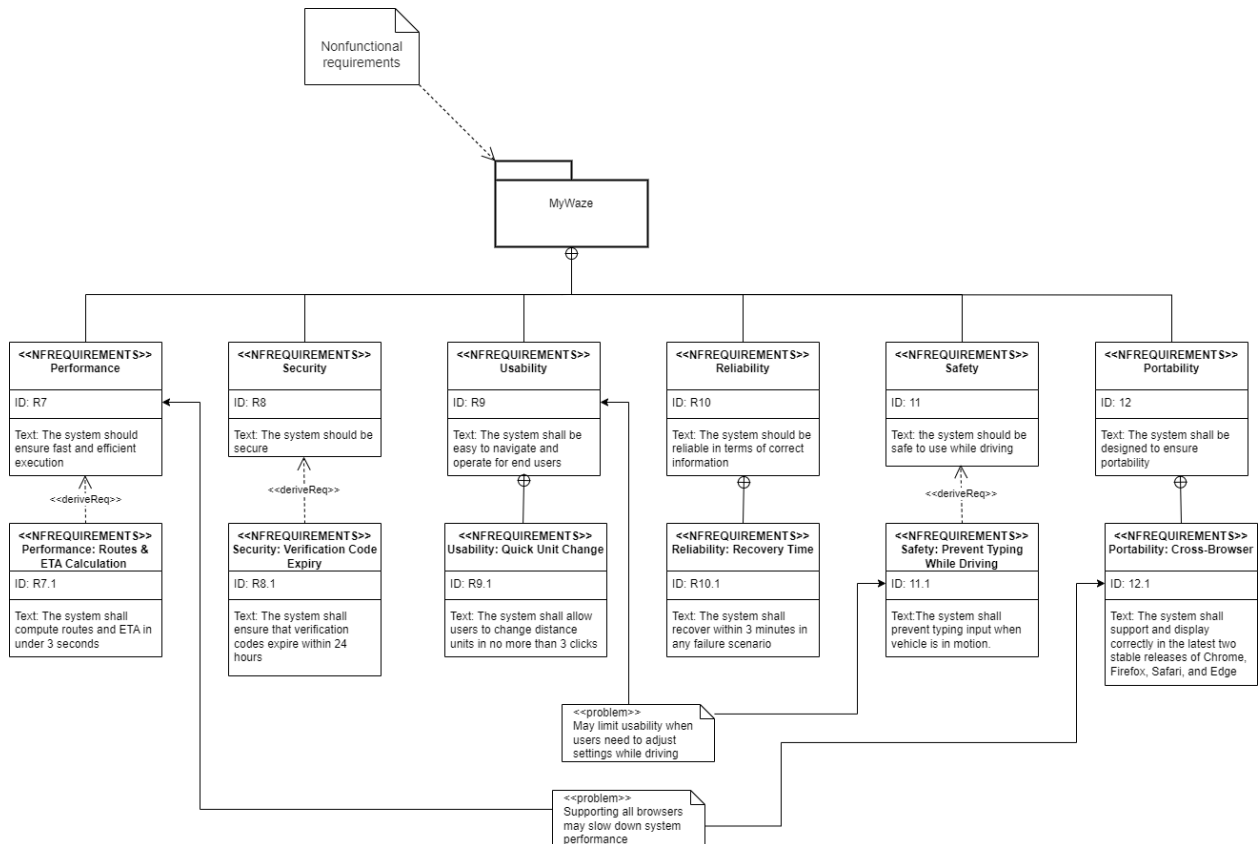
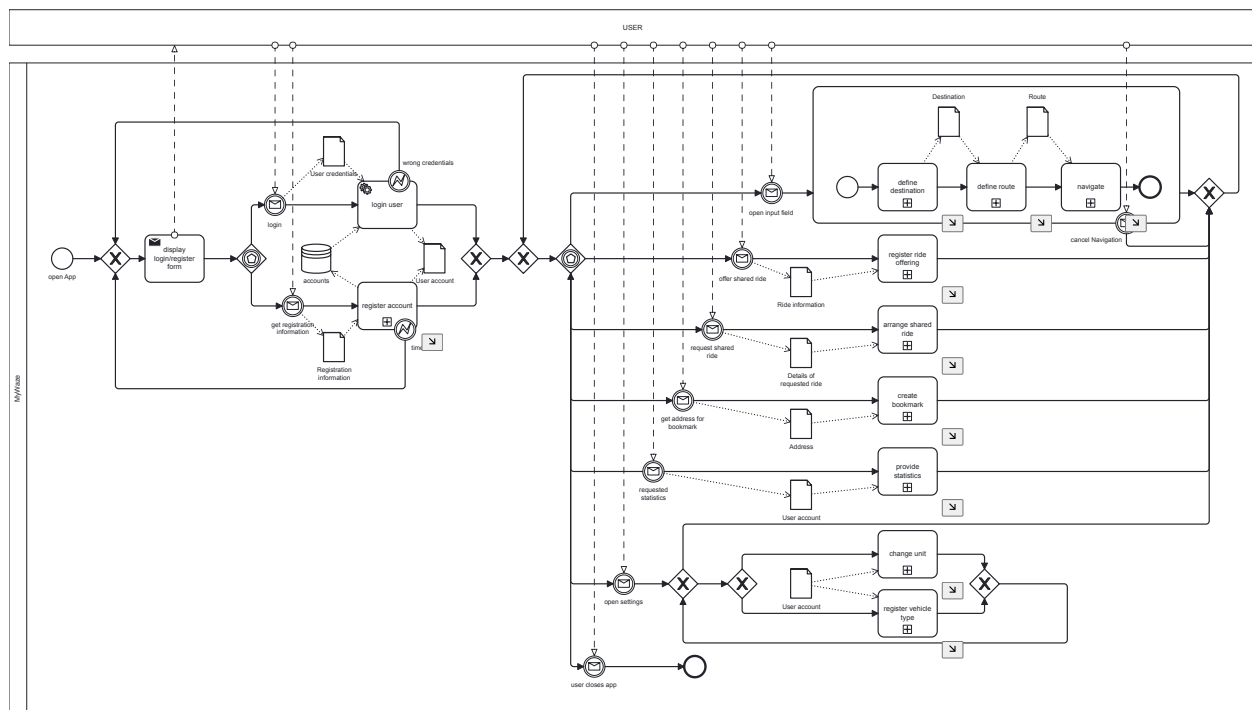


Figure 6: Nonfunctional Requirements

Mapping from Features to Models

Feature	Model
Overall process	MyWaze.bpmn
Register account	MyWaze_registerAccount.bpmn
Recent destinations	MyWaze_defineDestination.bpmn
Define route	MyWaze_defineRoute.bpmn, MyWaze_calculateRoute.bpmn
Avoid tolls	MyWaze_defineRoute.bpmn
Walking route	MyWaze_defineRoute.bpmn
ETA	MyWaze_defineRoute.bpmn
Current speed	MyWaze_navigate.bpmn, MyWaze_speedMonitoring.bpmn
Speed alert	MyWaze_navigate.bpmn, MyWaze_speedMonitoring.bpmn
Find parking lots	MyWaze_navigate.bpmn, MyWaze_defineRouteToParkingLot.bpmn, MyWaze_calculateRoute.bpmn
Hazard reports	MyWaze_navigate.bpmn, MyWaze_reportHazard.bpmn
Share ride	MyWaze.bpmn, MyWaze_registerRideOffering.bpmn, MyWaze_arrangeSharedRide.bpmn
Bookmarking	MyWaze.bpmn, MyWaze_createBookmark.bpmn
Statistics	MyWaze.bpmn, MyWaze_provideStatistics.bpmn
Change units	MyWaze.bpmn, MyWaze_changeUnits.bpmn
Register vehicle type	MyWaze.bpmn, MyWaze_registerVehicleType.bpmn

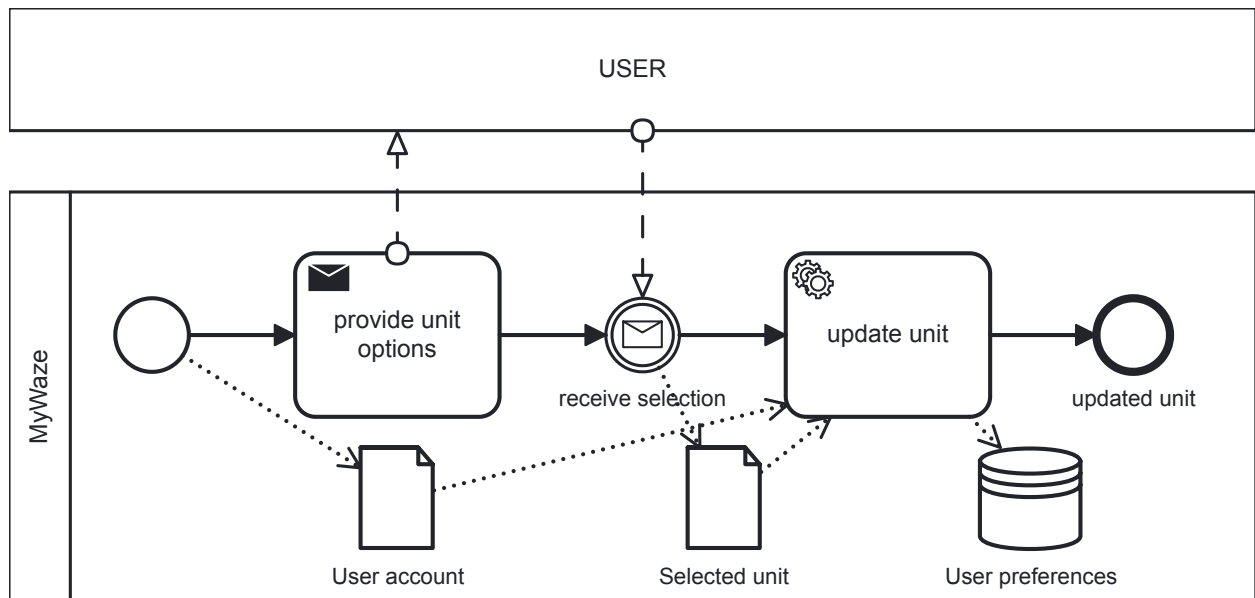
BPMN Models



MyWaze.bpmn

BPMN.IO

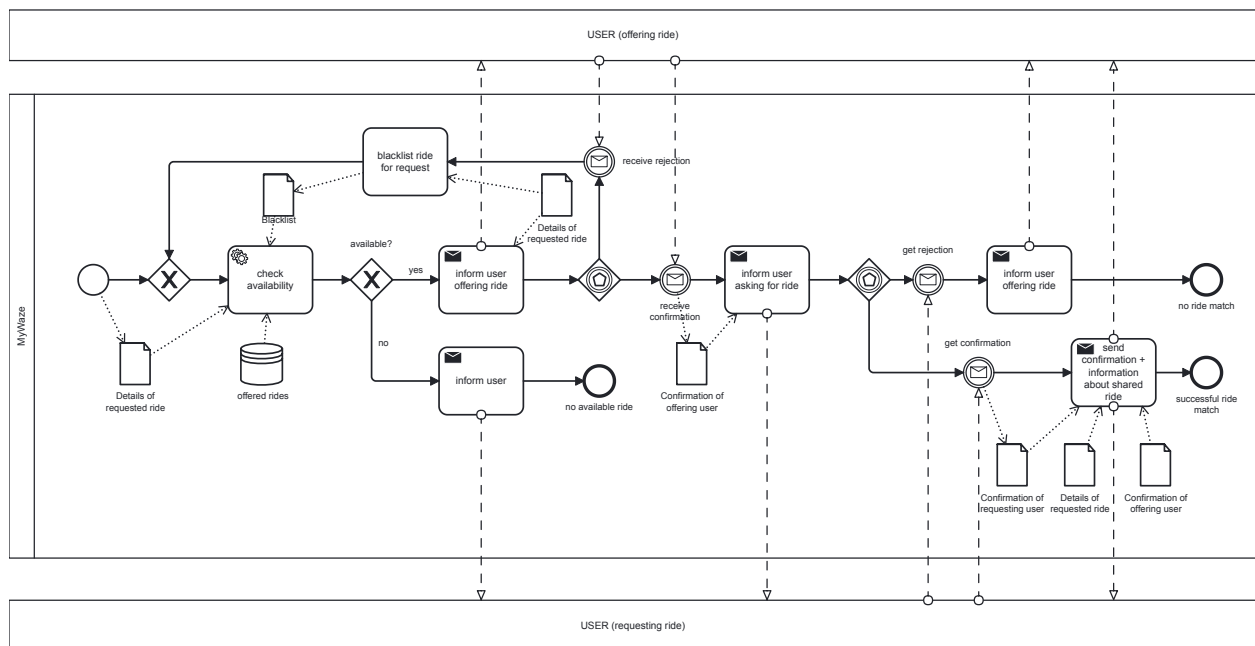
Figure 7: Model: MyWaze



MyWaze_ChangeUnit.bpmn

BPMN.io

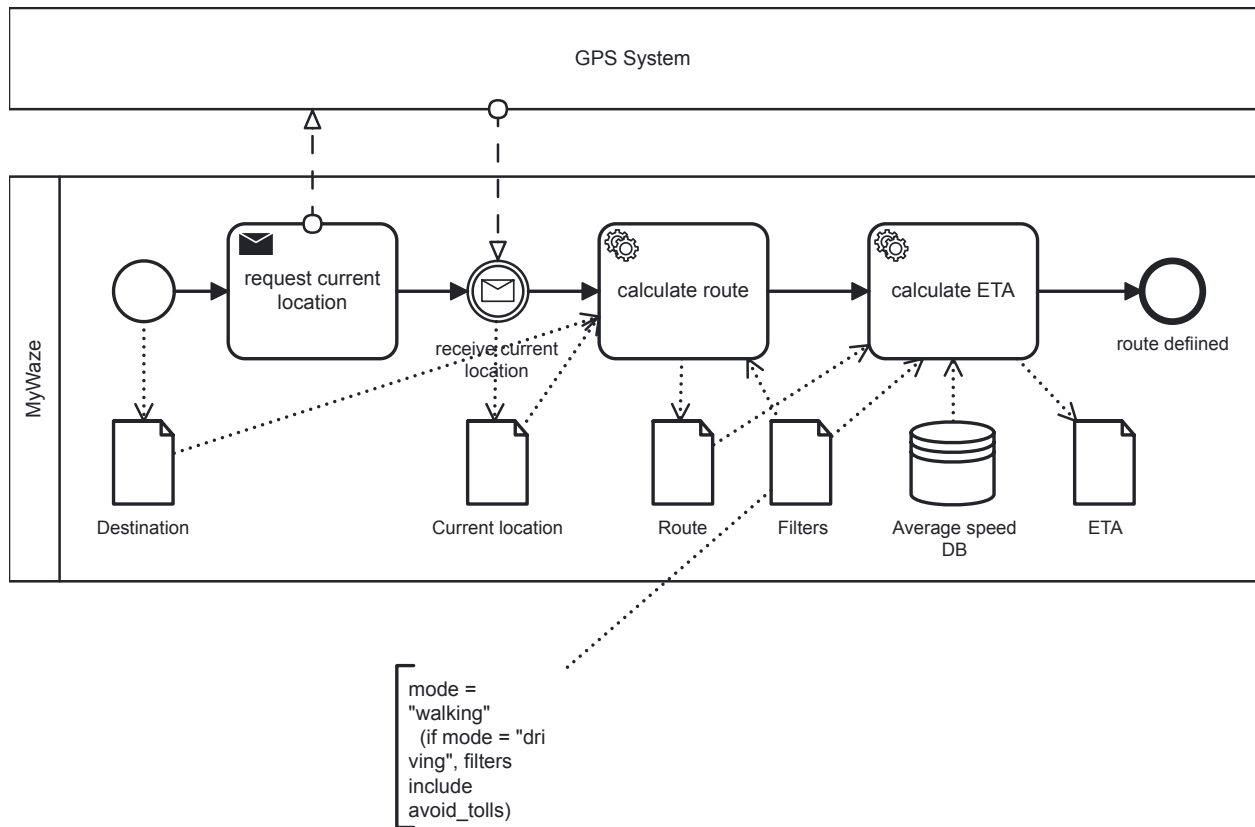
Figure 8: Model: MyWaze'ChangeUnit



MyWaze_arrangeSharedRide.bpmn

BPMN.io

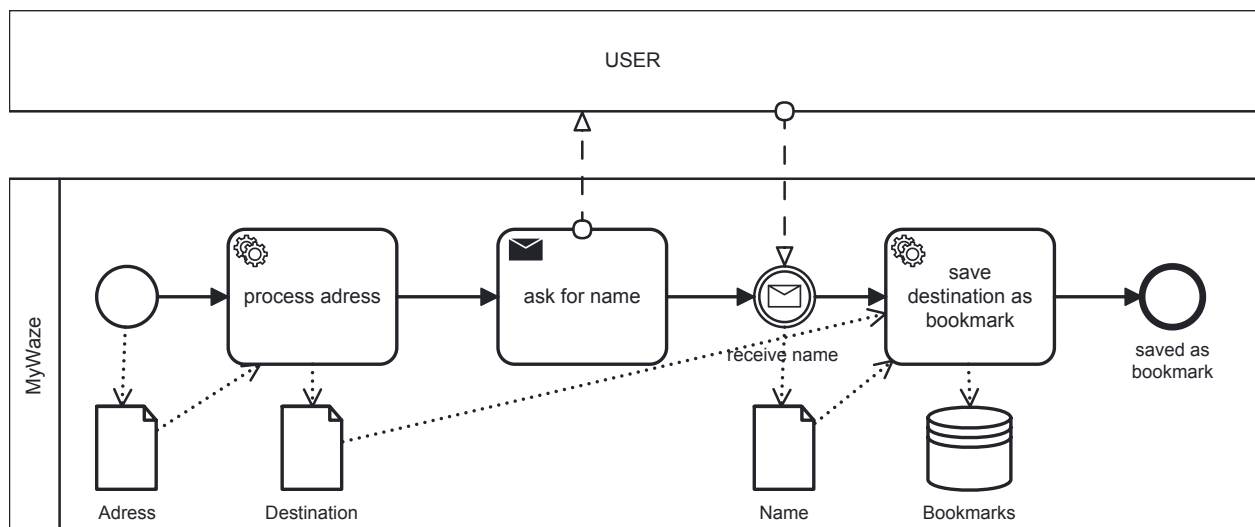
Figure 9: Model: MyWaze'arrangeSharedRide



MyWaze_calculateRoute.bpmn

BPMN.iO

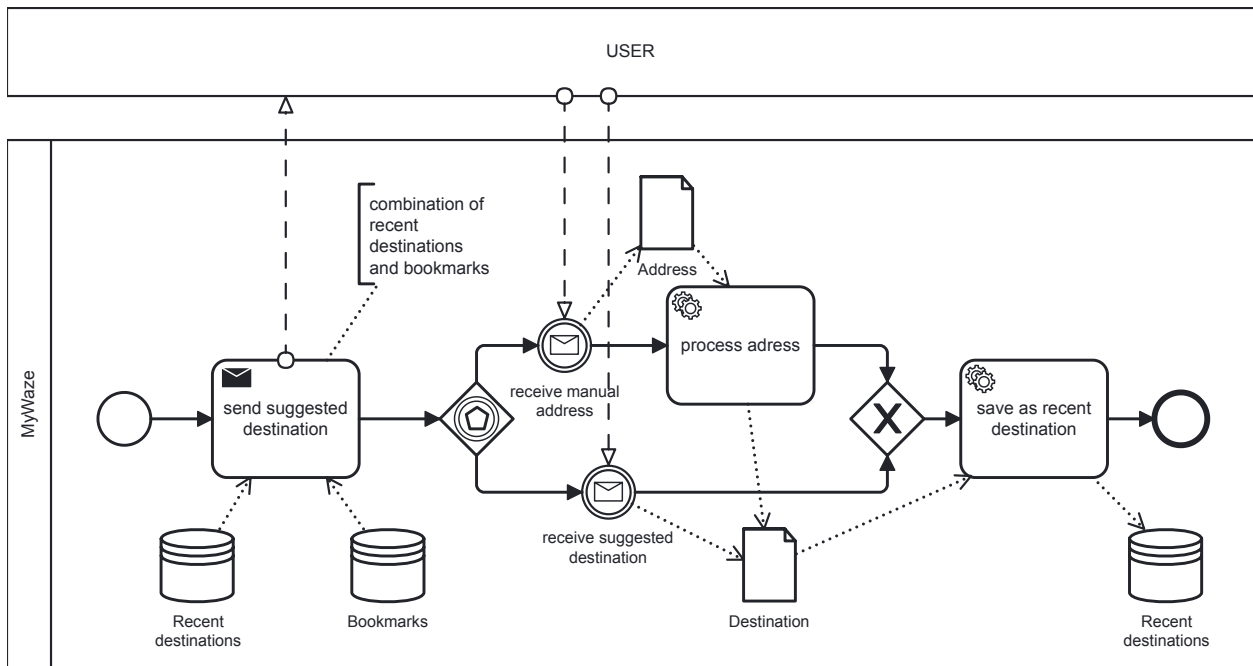
Figure 10: Model: MyWaze'calculateRoute



MyWaze_createBookmark.bpmn

BPMN.iO

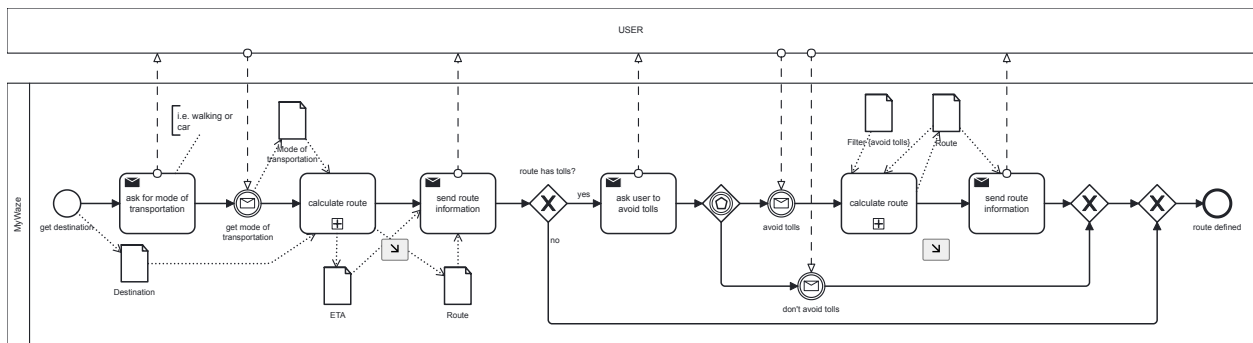
Figure 11: Model: MyWaze'createBookmark



MyWaze_defineDestination.bpmn

BPMN.io

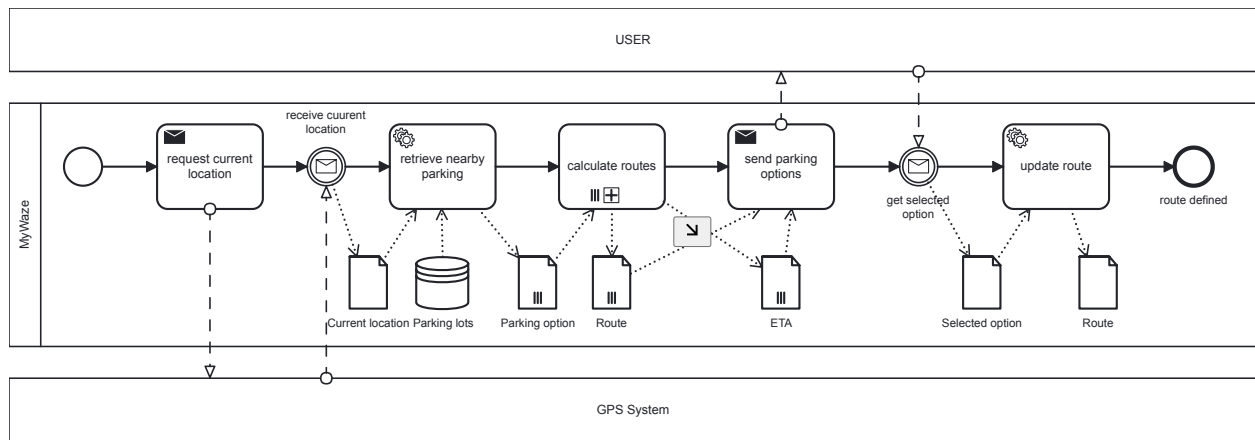
Figure 12: Model: MyWaze 'defineDestination'



MyWaze_defineRoute.bpmn

BPMN.io

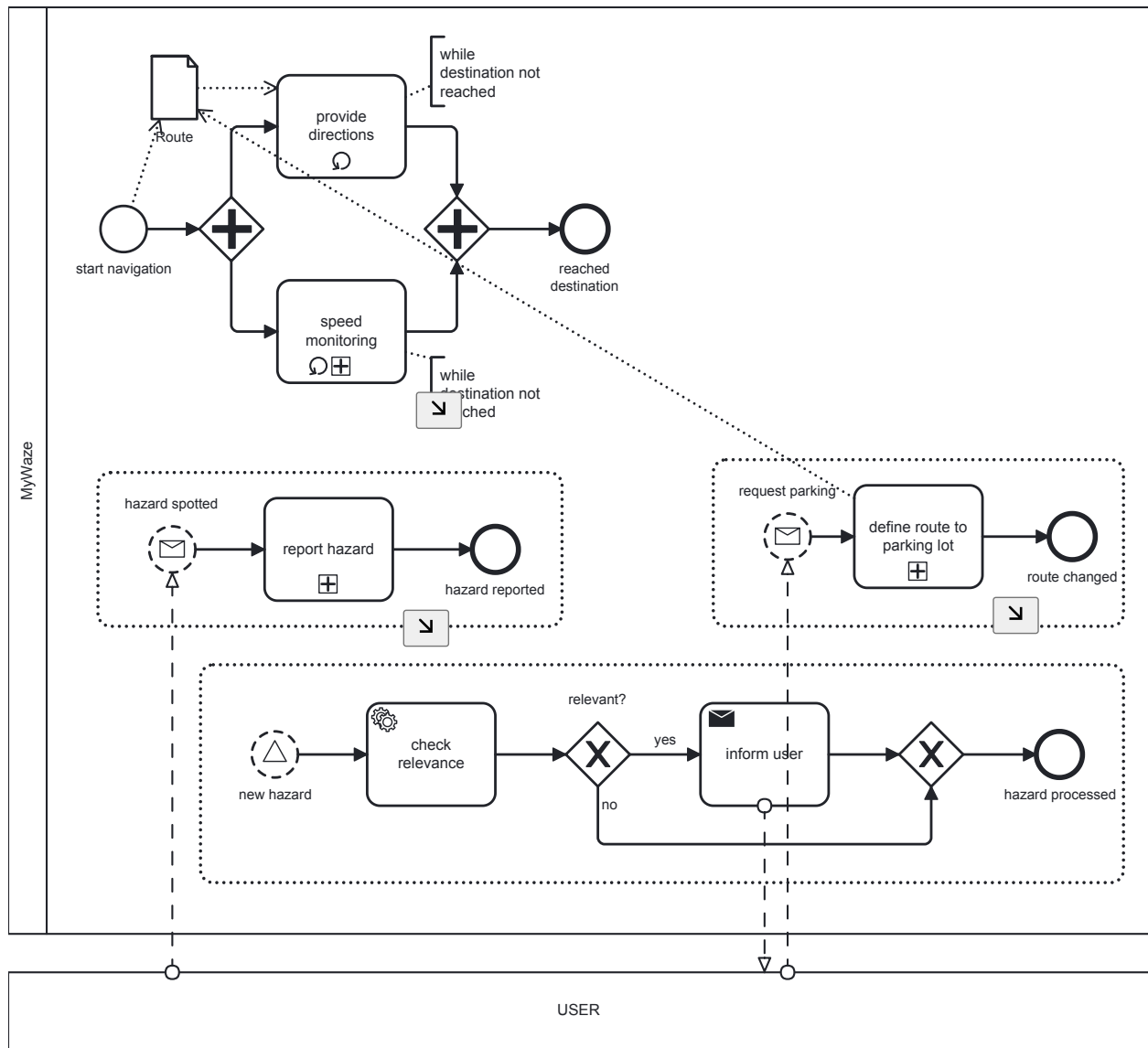
Figure 13: Model: MyWaze 'defineRoute'



MyWaze_defineRouteToParkingLot.bpmn

BPMN.iO

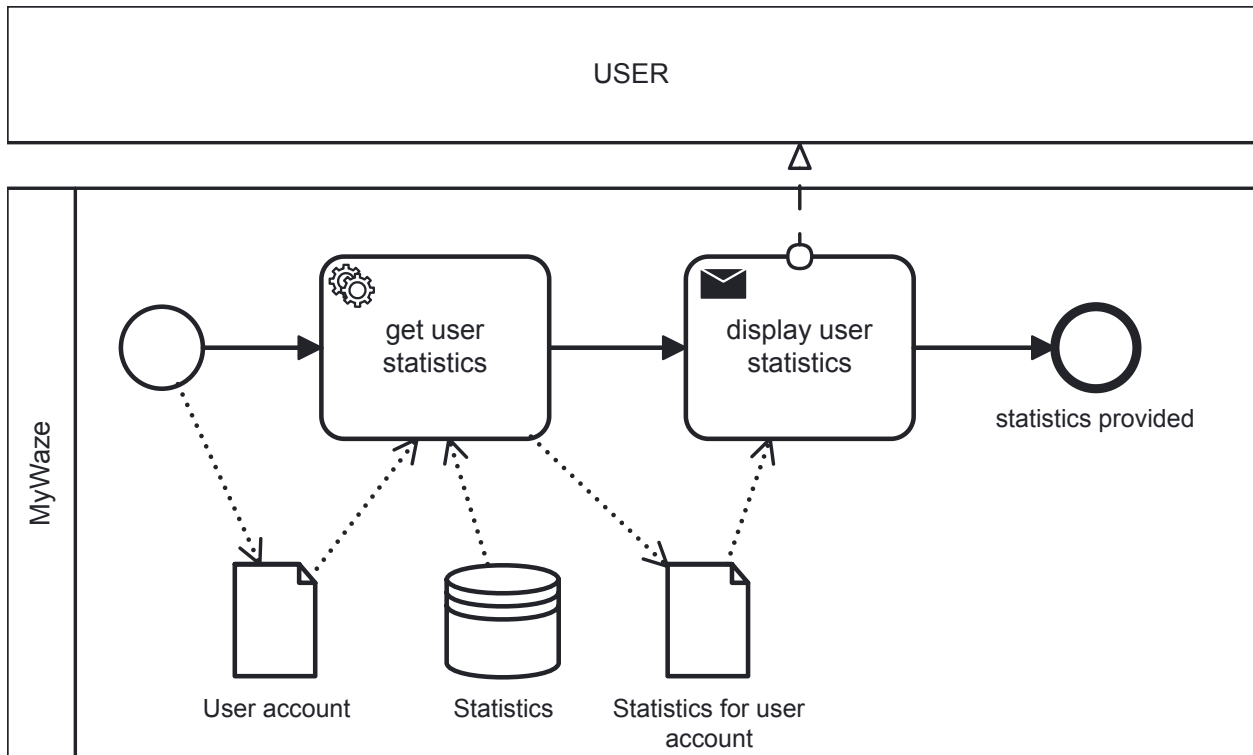
Figure 14: Model: MyWaze defineRouteToParkingLot



MyWaze_navigate.bpmn

BPMN.IO

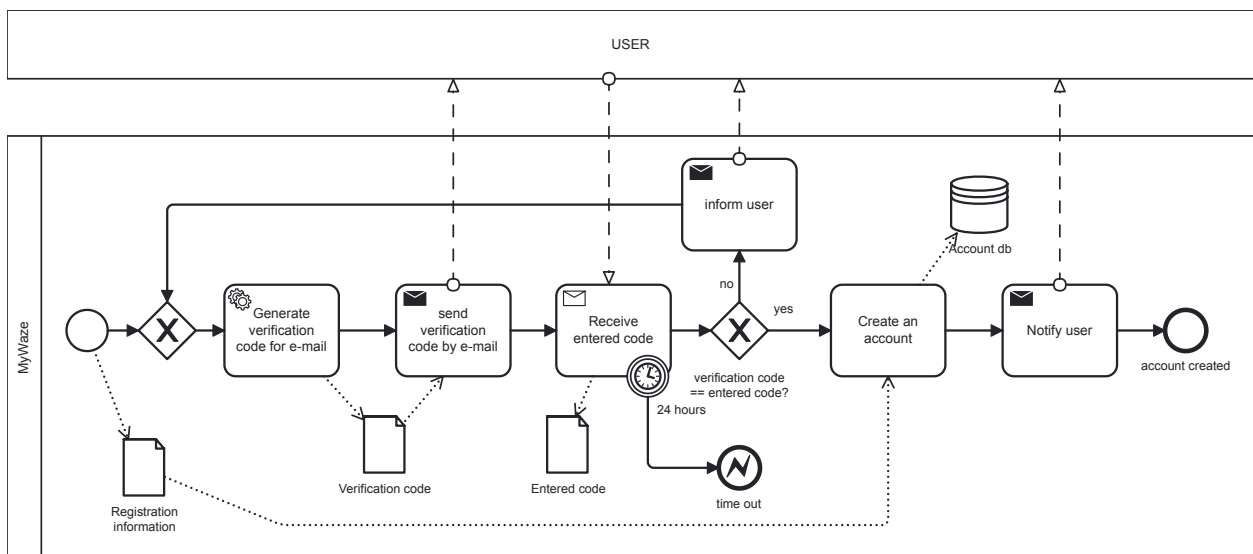
Figure 15: Model: MyWaze navigate



MyWaze_provideStatistics.bpmn

BPMN.io

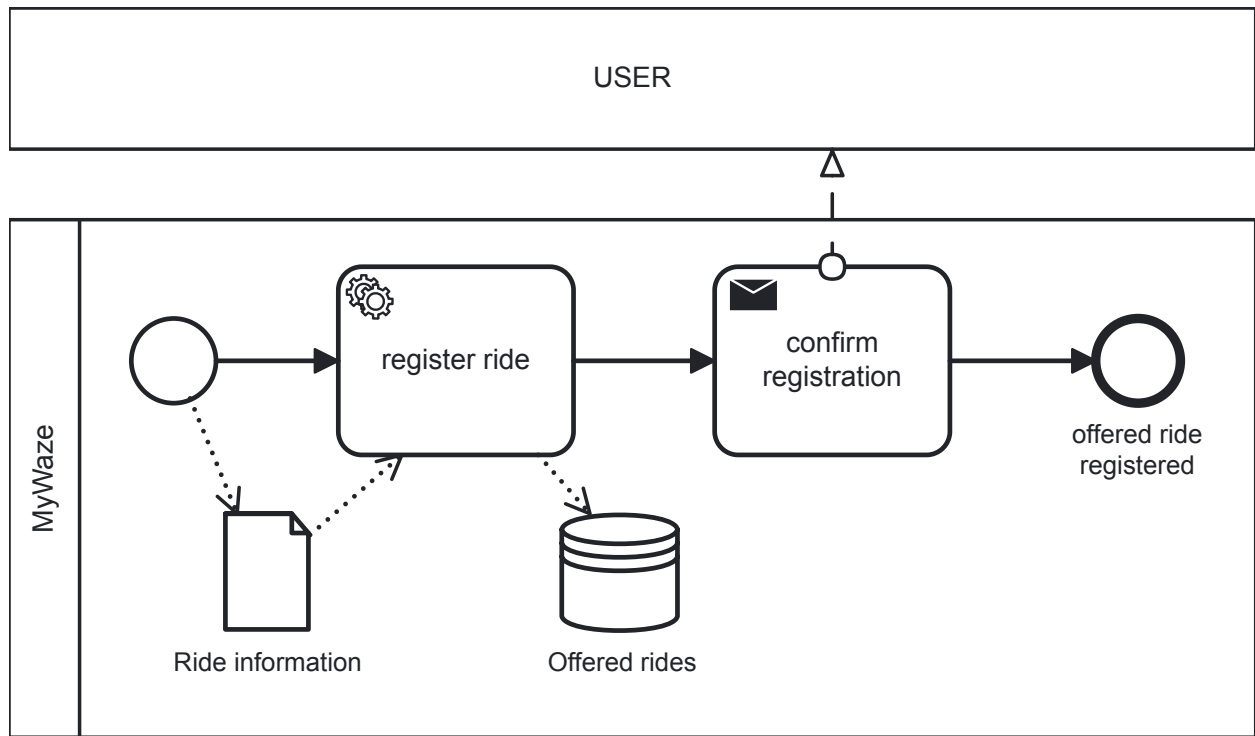
Figure 16: Model: MyWaze`provideStatistics



MyWaze_registerAccount.bpmn

BPMN.io

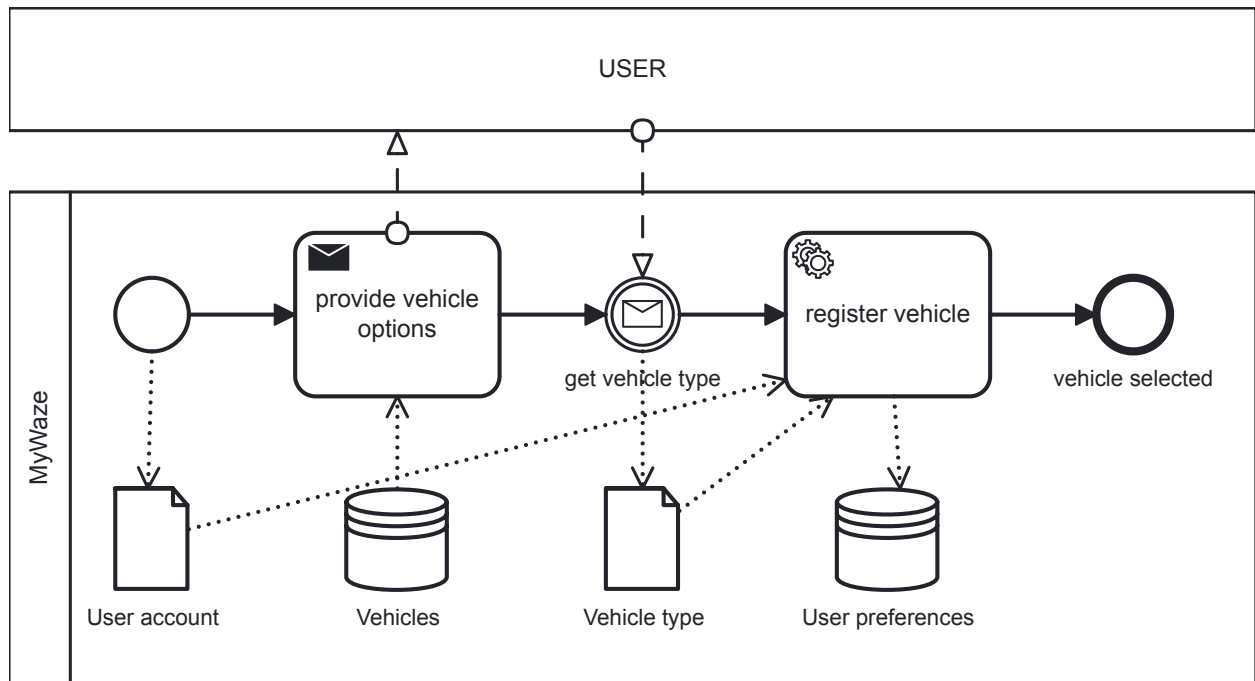
Figure 17: Model: MyWaze`registerAccount



MyWaze_registerRideOffering.bpmn

BPMN.iO

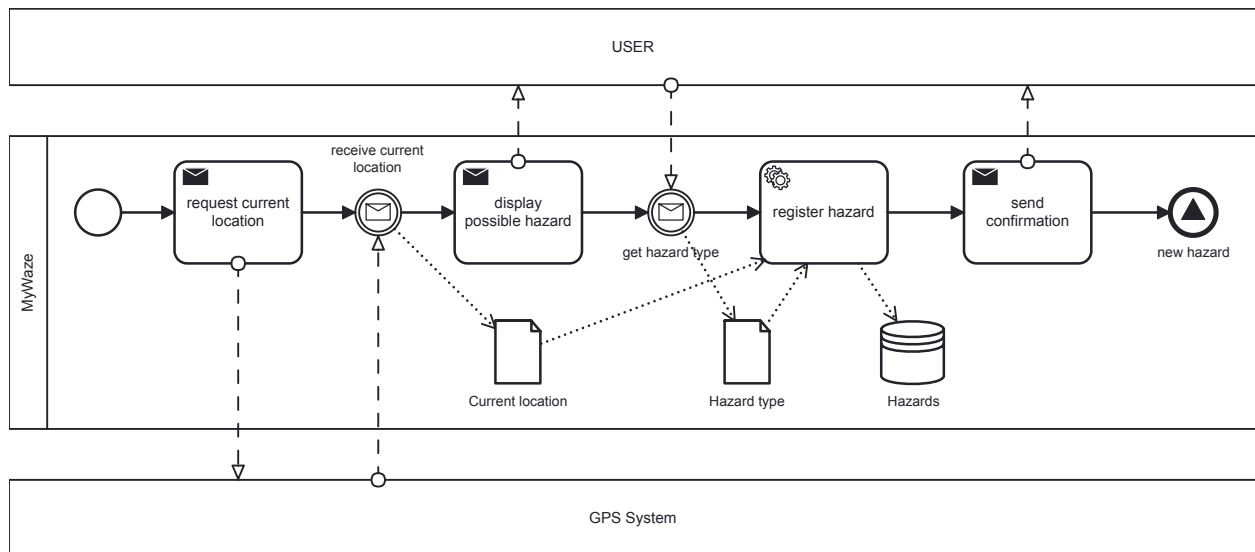
Figure 18: Model: MyWaze`registerRideOffering



MyWaze_registerVehicleType.bpmn

BPMN.iO

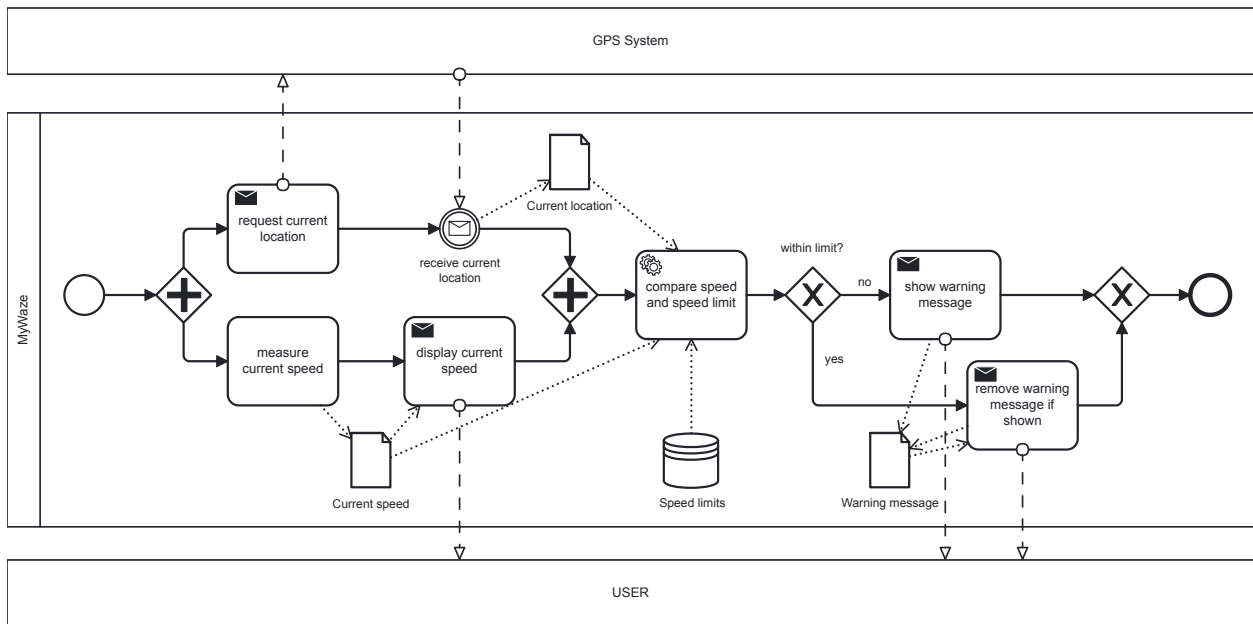
Figure 19: Model: MyWaze'registerVehicleType



MyWaze_reportHazard.bpmn

BPMN.iO

Figure 20: Model: MyWaze'reportHazard



MyWaze_speedMonitoring.bpmn

BPMN.io

Figure 21: Model: MyWaze's speedMonitoring

Context Diagram

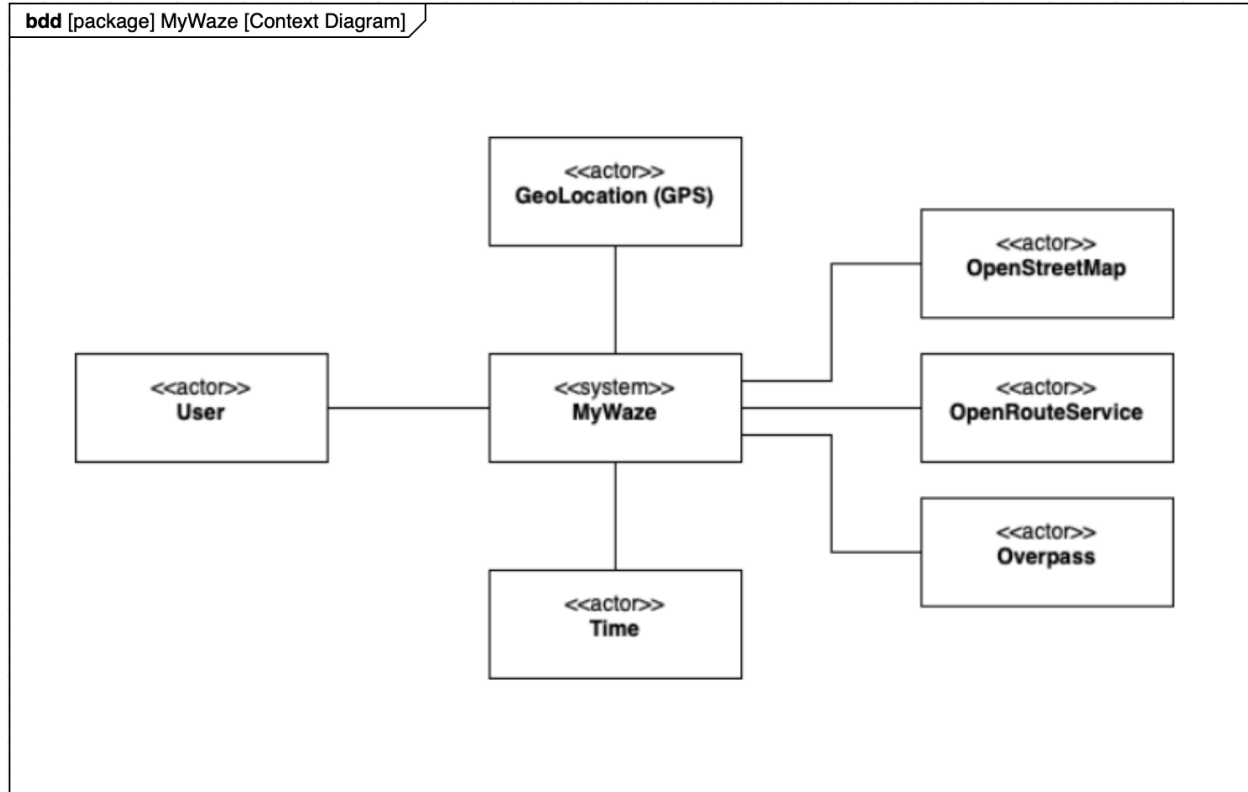


Figure 22: Context Diagram

Use Case View

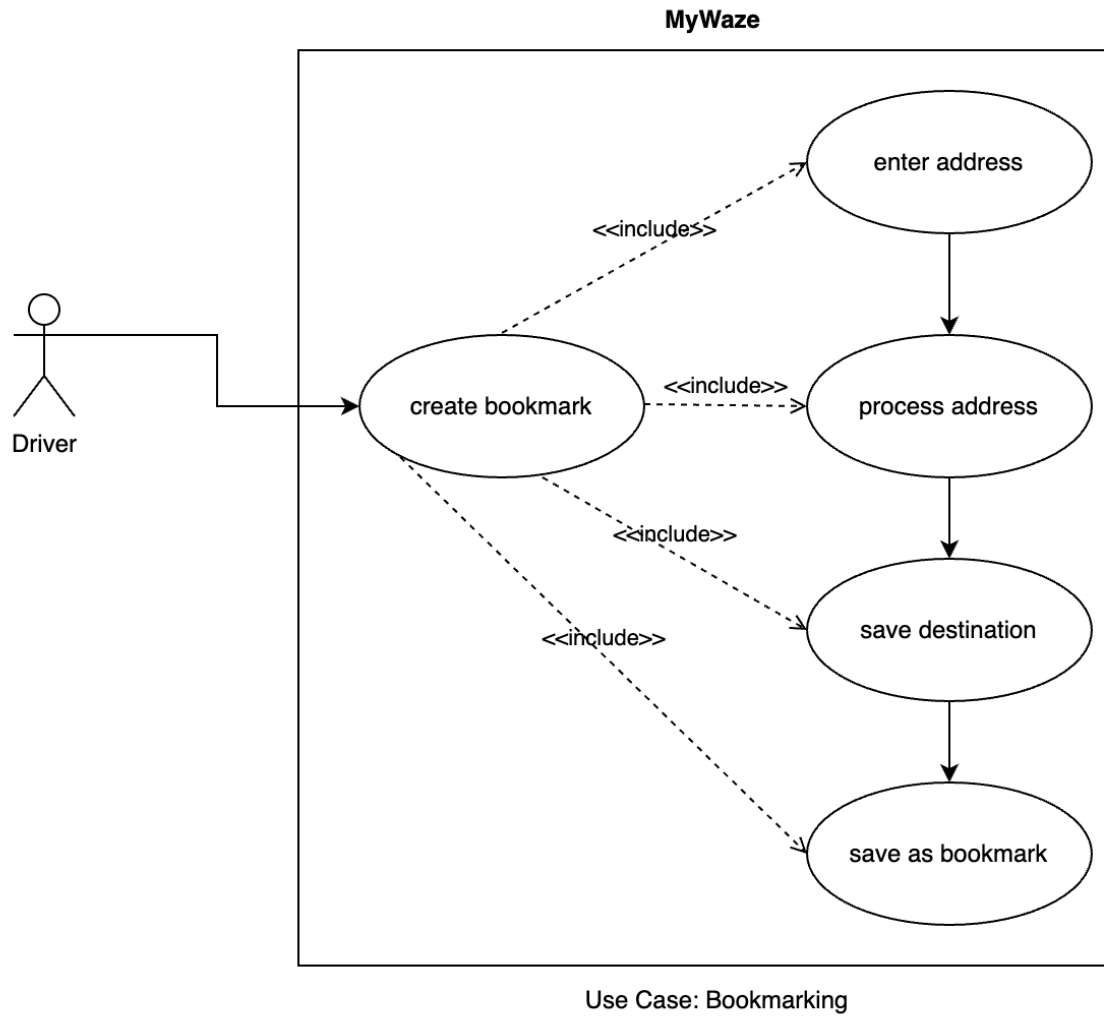
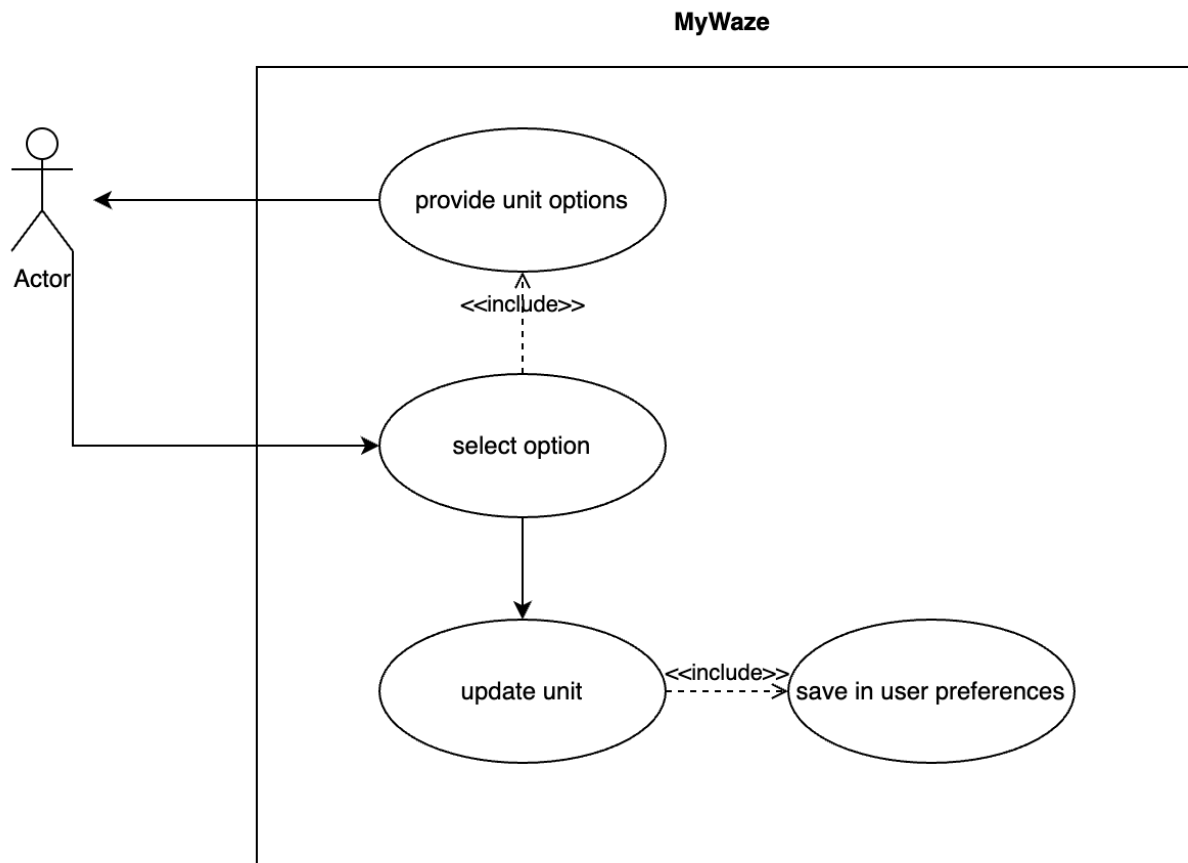


Figure 23: Use Case Diagram: Bookmarking



Use Case Diagram: Change Units

Figure 24: Use Case Diagram: Change Units

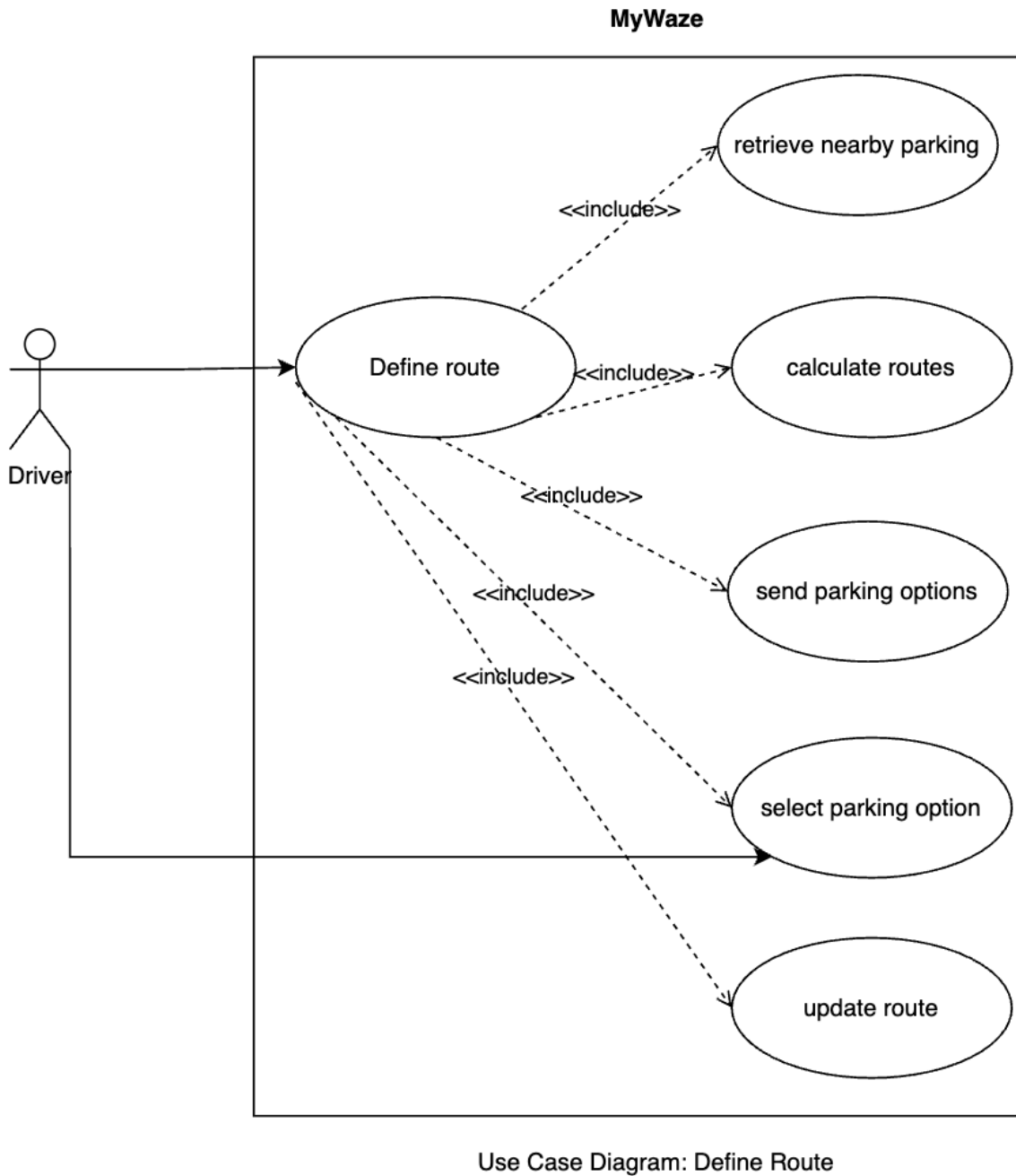


Figure 25: Use Case Diagram: Define Route

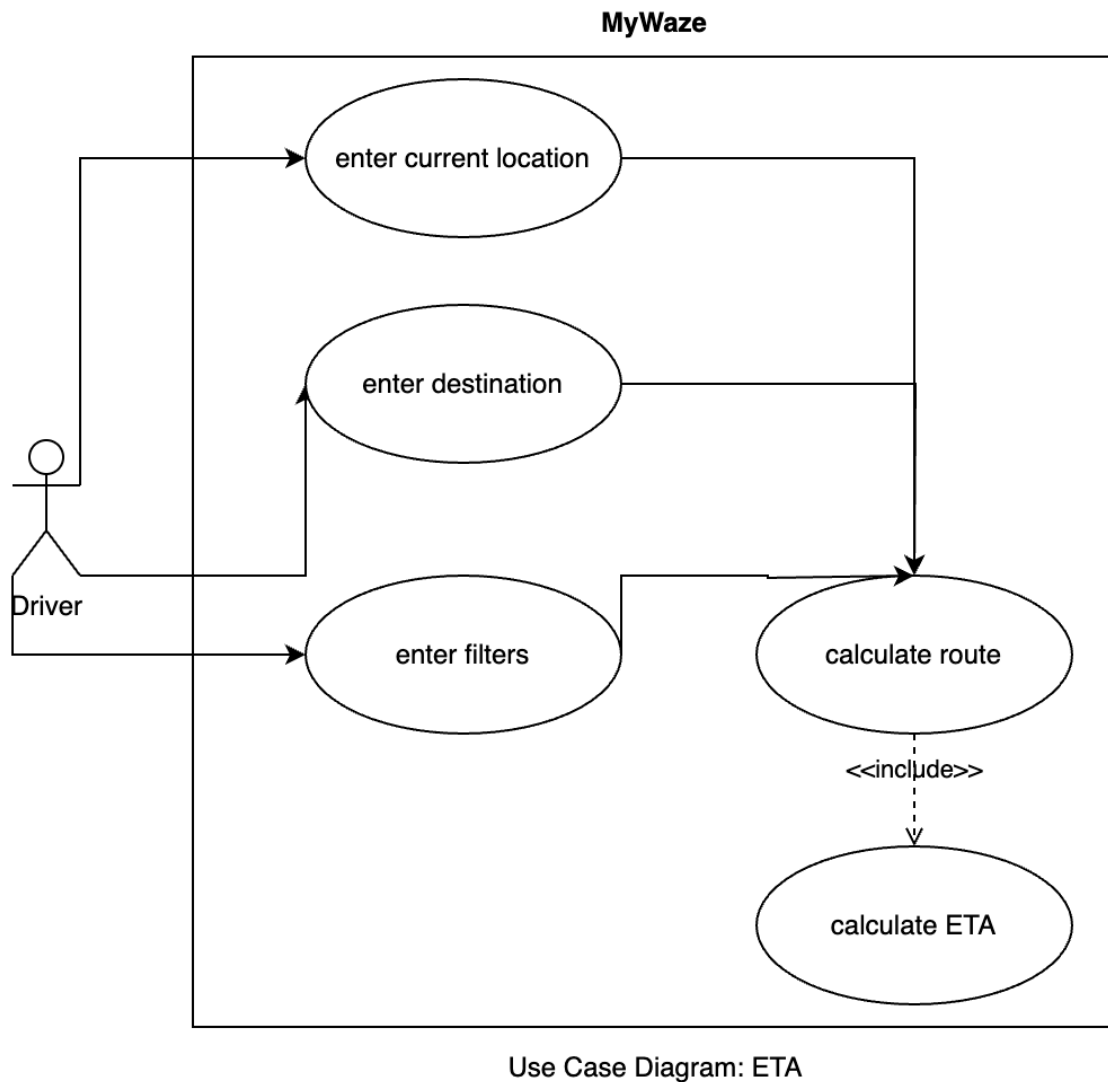


Figure 26: Use Case Diagram: ETA

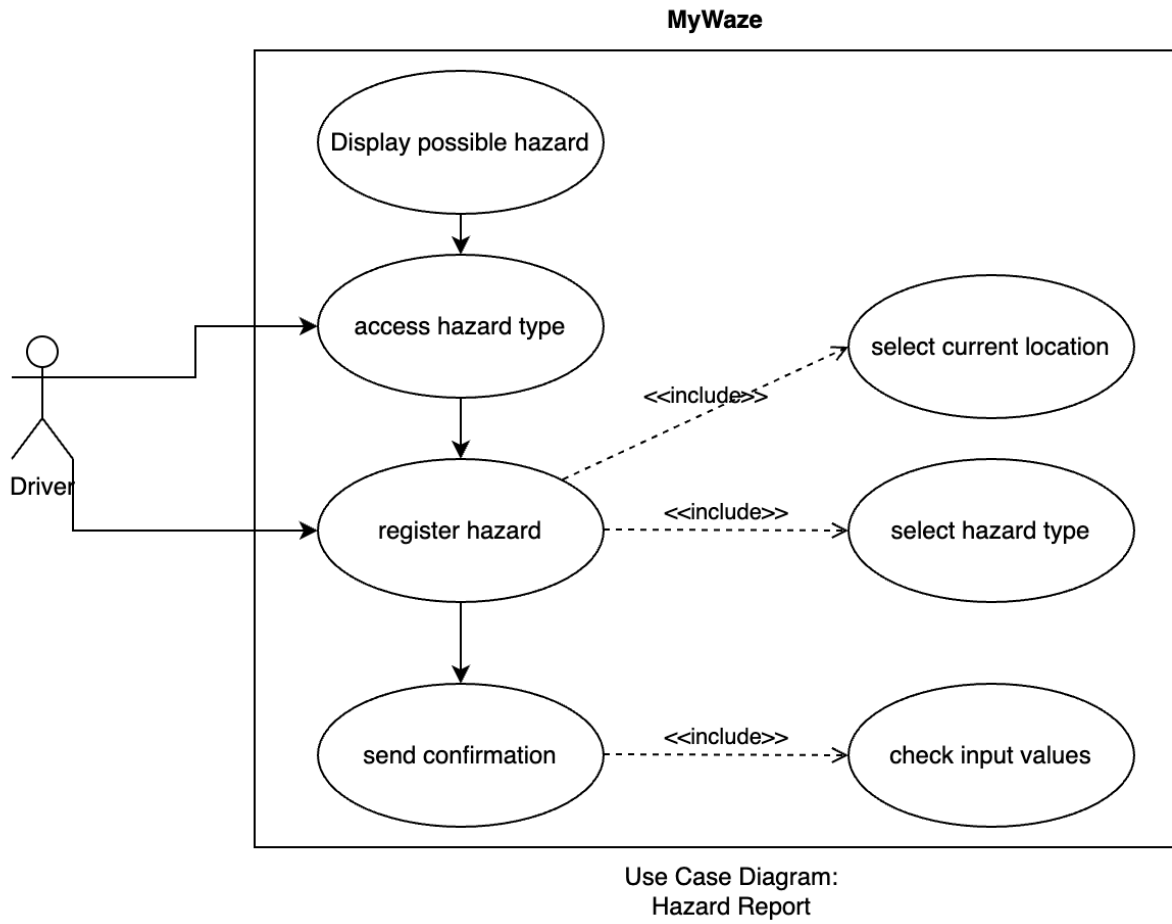
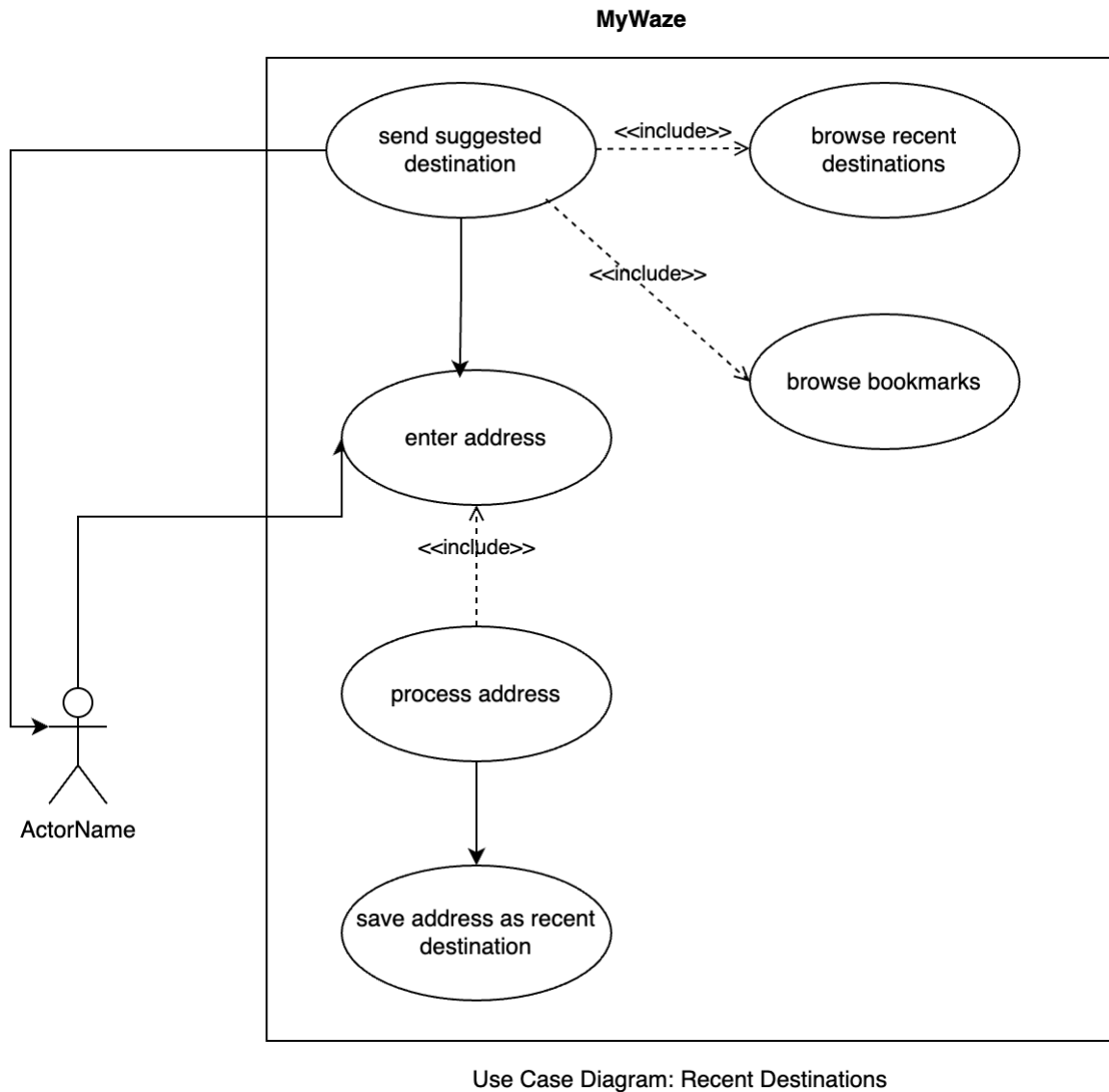


Figure 27: Use Case Diagram: Hazard Report



Use Case Diagram: Recent Destinations

Figure 28: Use Case Diagram: Recent Destinations

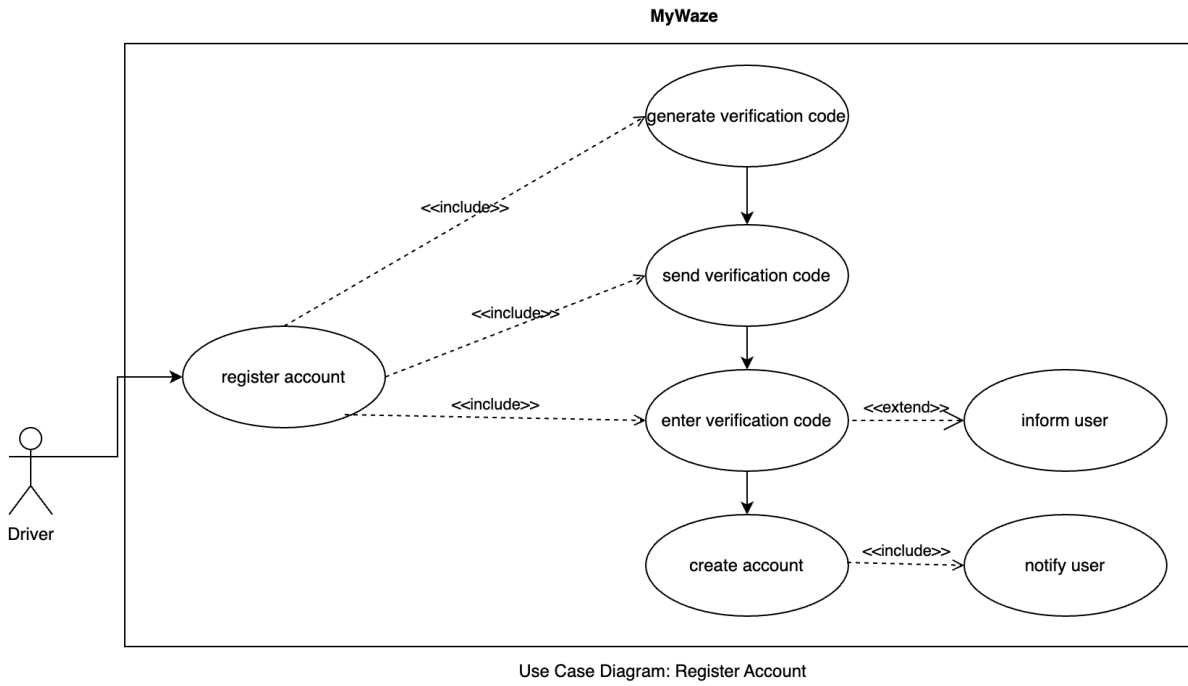


Figure 29: Use Case Diagram: Register Account

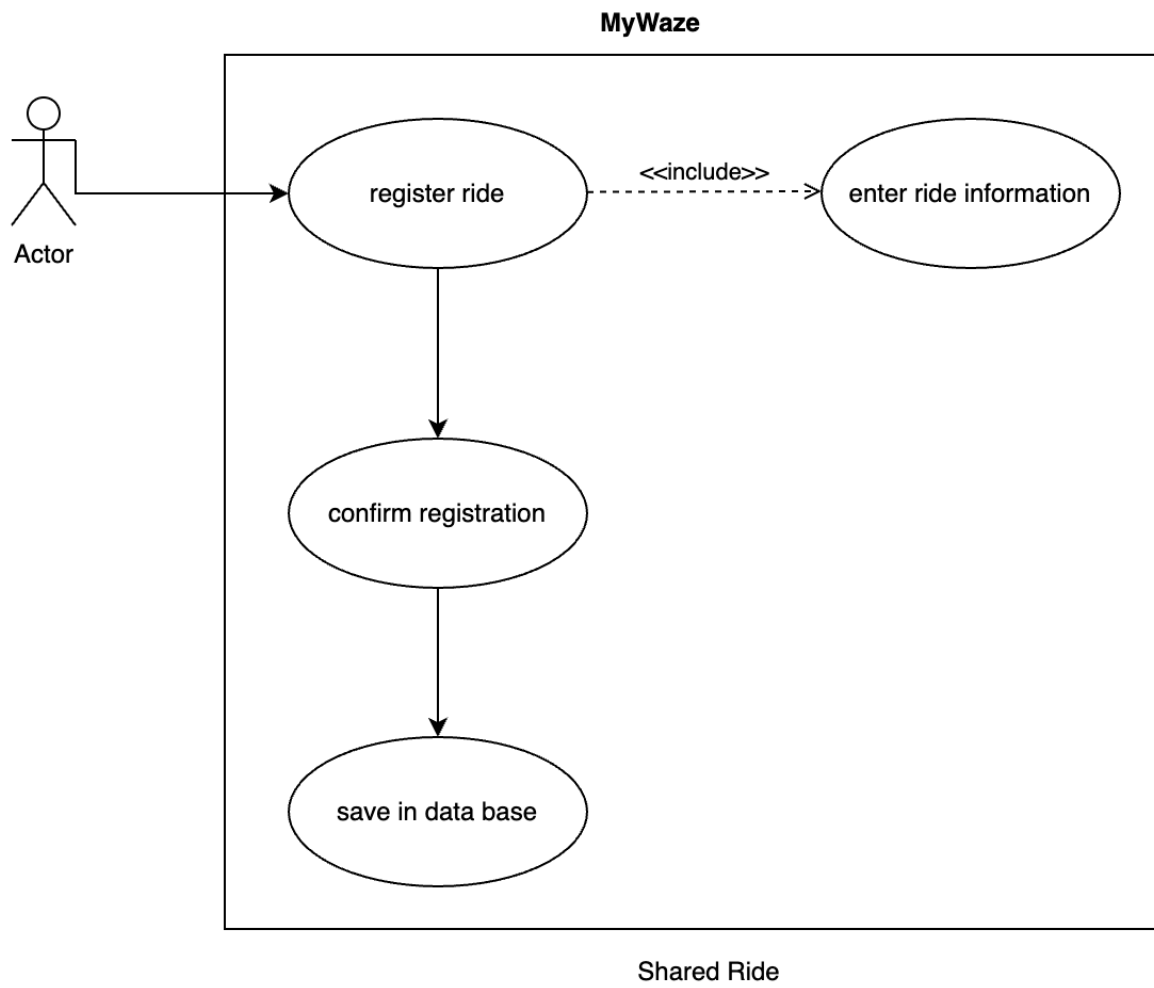


Figure 30: Use Case Diagram: Shared Ride

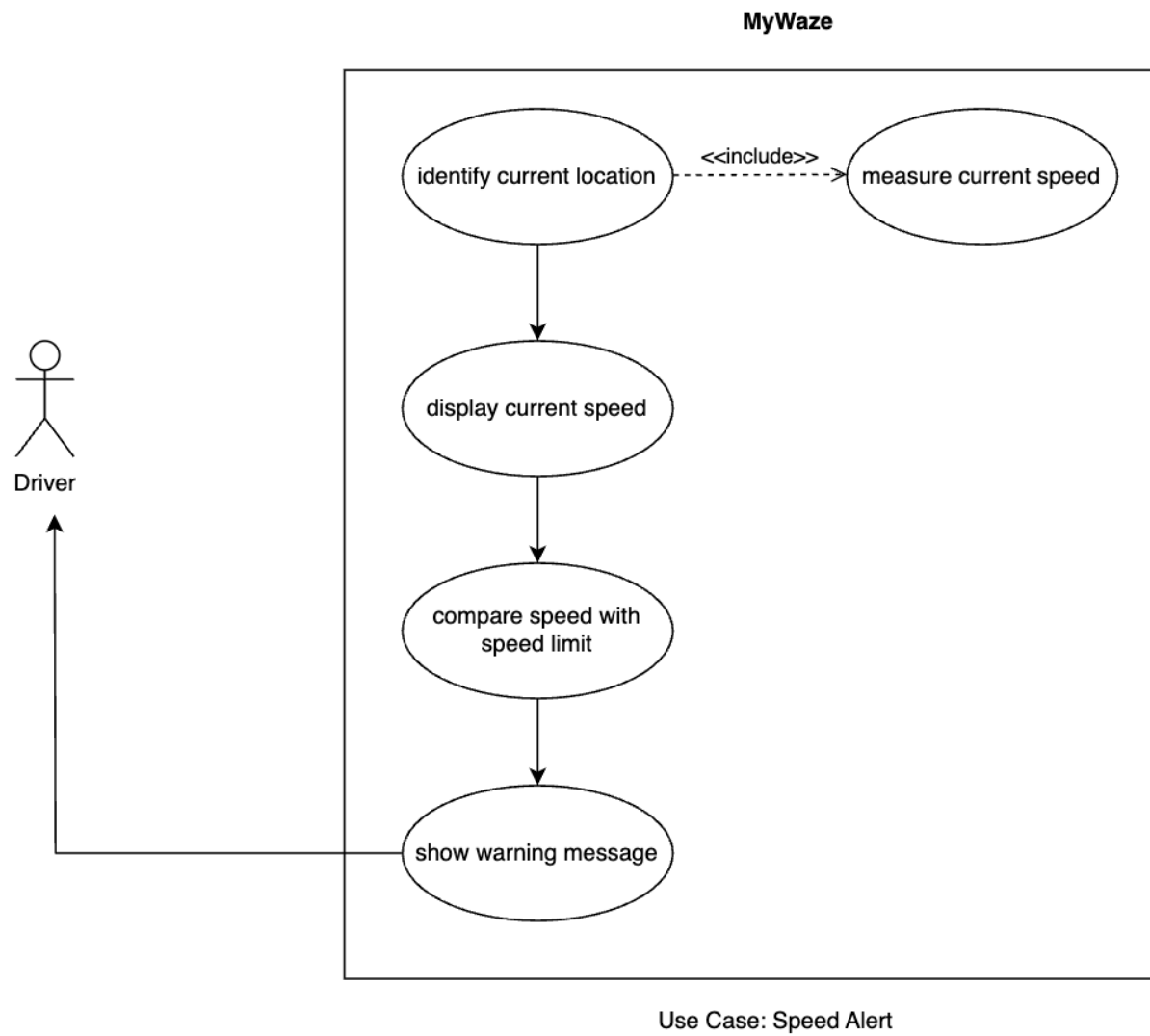


Figure 31: Use Case Diagram: Speed Alert

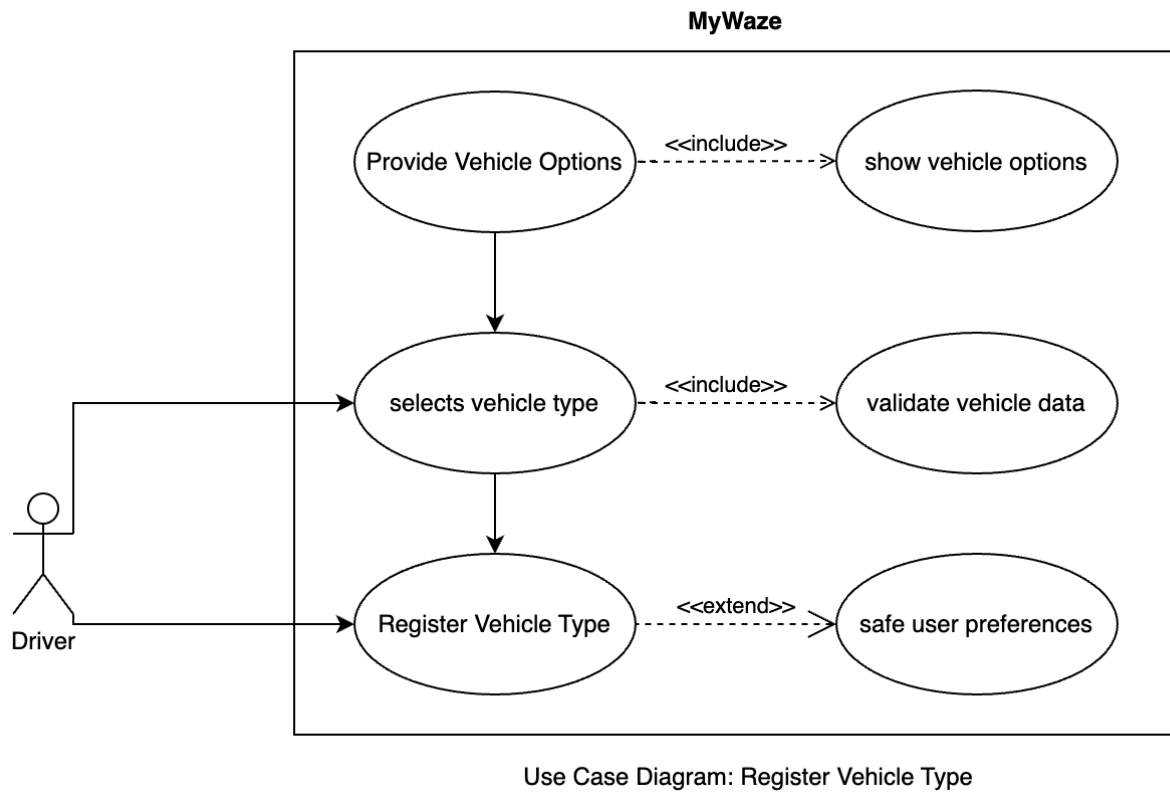


Figure 32: Use Case Diagram: Register Vehicle Type

Logical View

System Decomposition

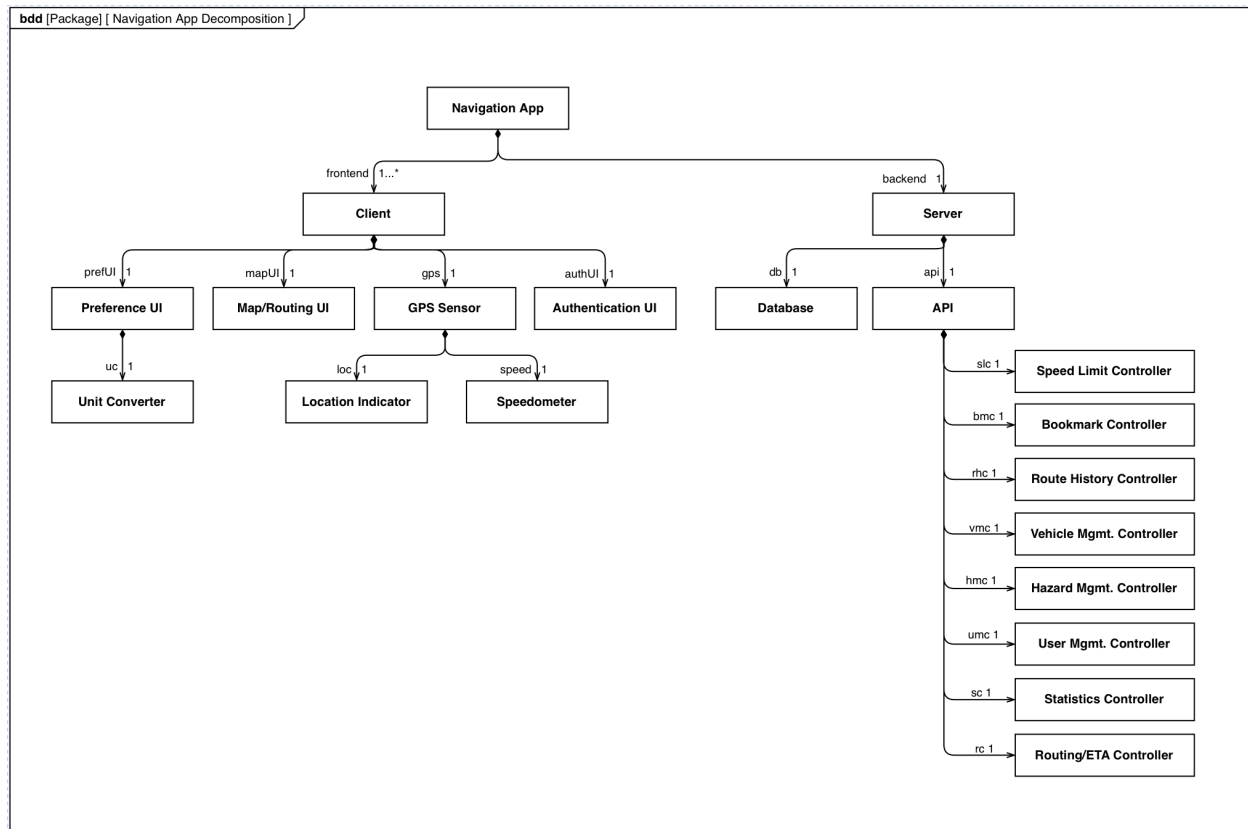


Figure 33: MyWaze Decomposition

Navigation Domain

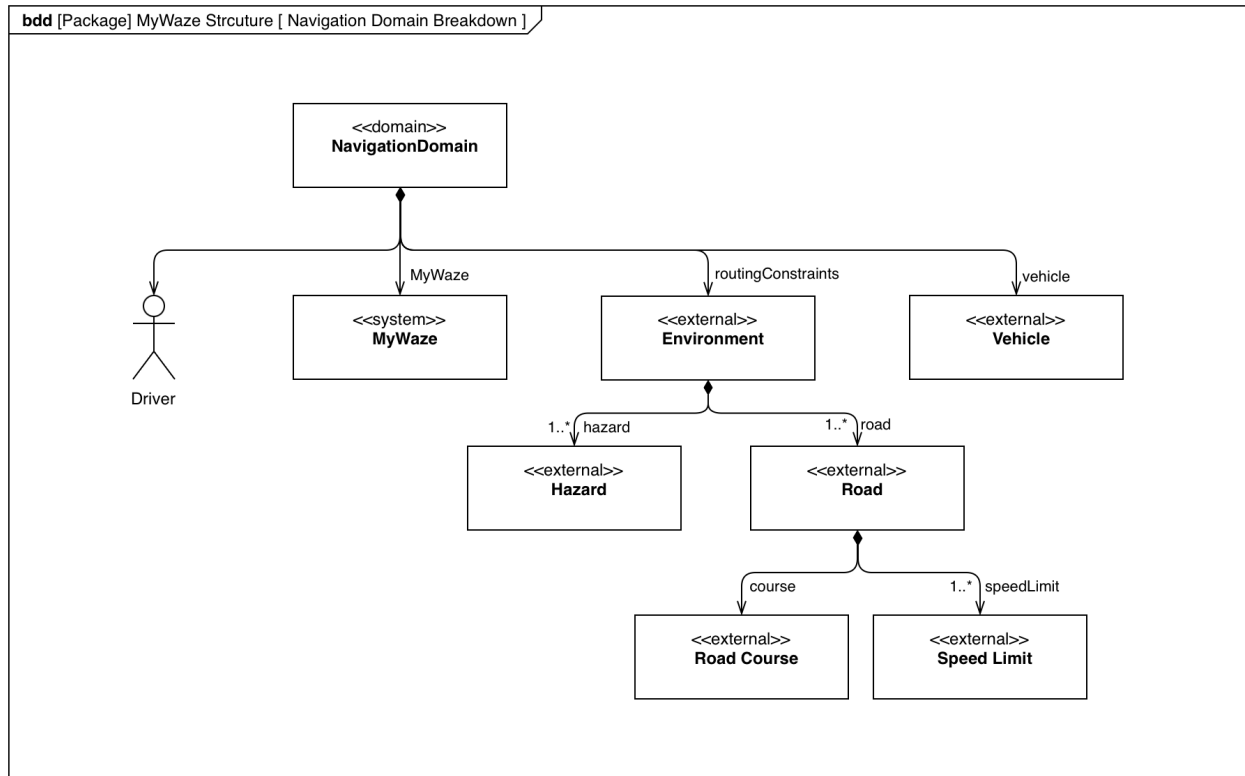


Figure 34: Navigation Domain

Process View

Sequence Diagramm - Select Recent Destination

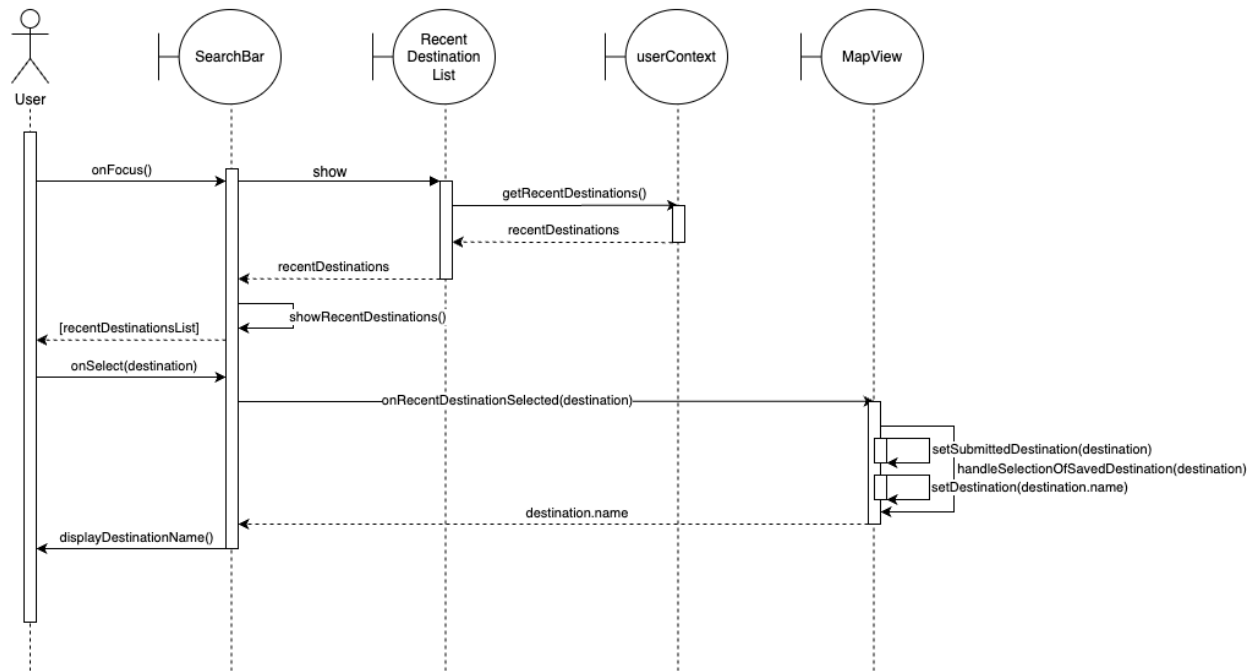


Figure 35: Sequence Diagram: Recent Destinations

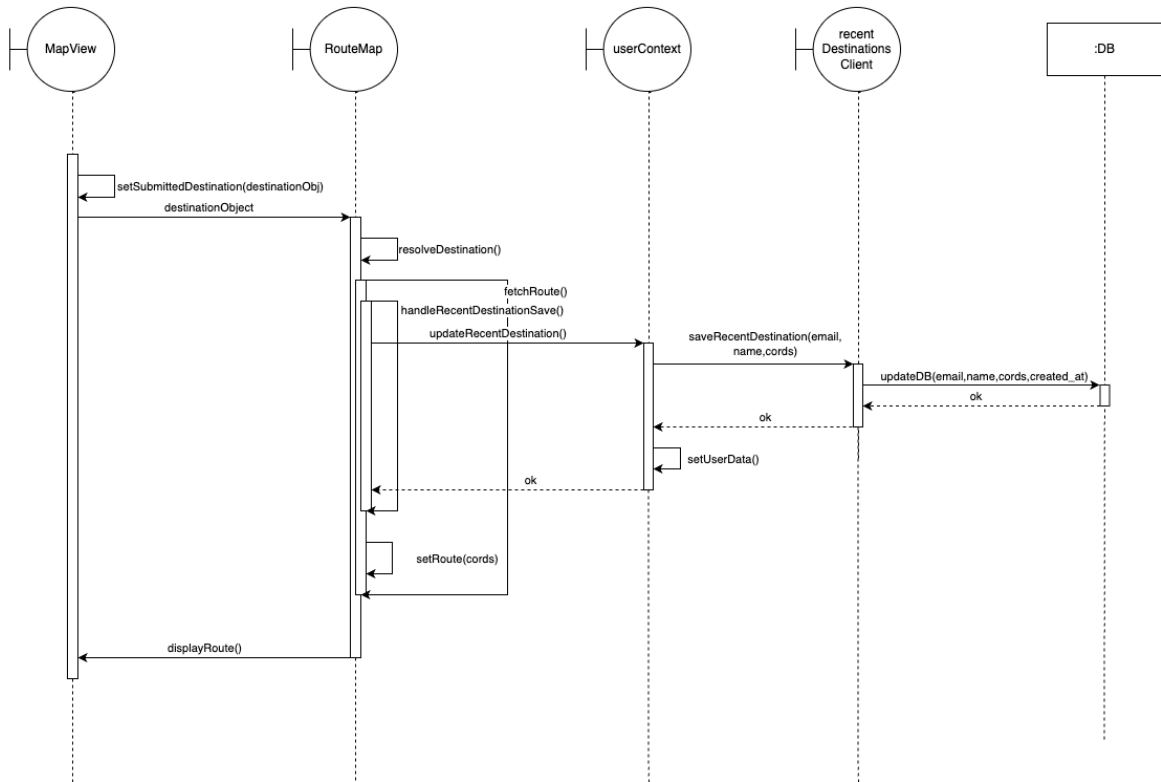


Figure 36: Sequence Diagram: Routing

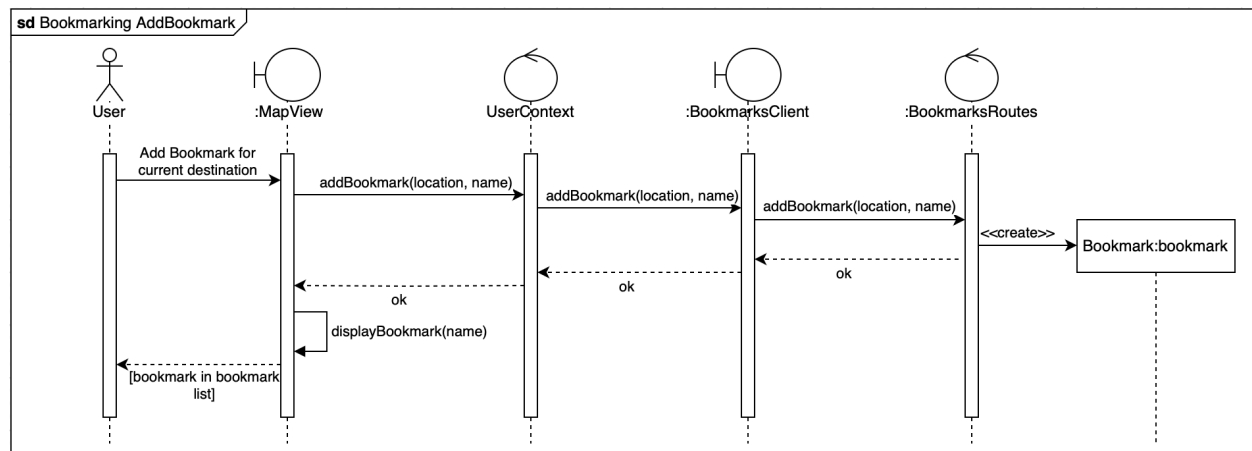


Figure 37: Sequence Diagram: Bookmarking (add)

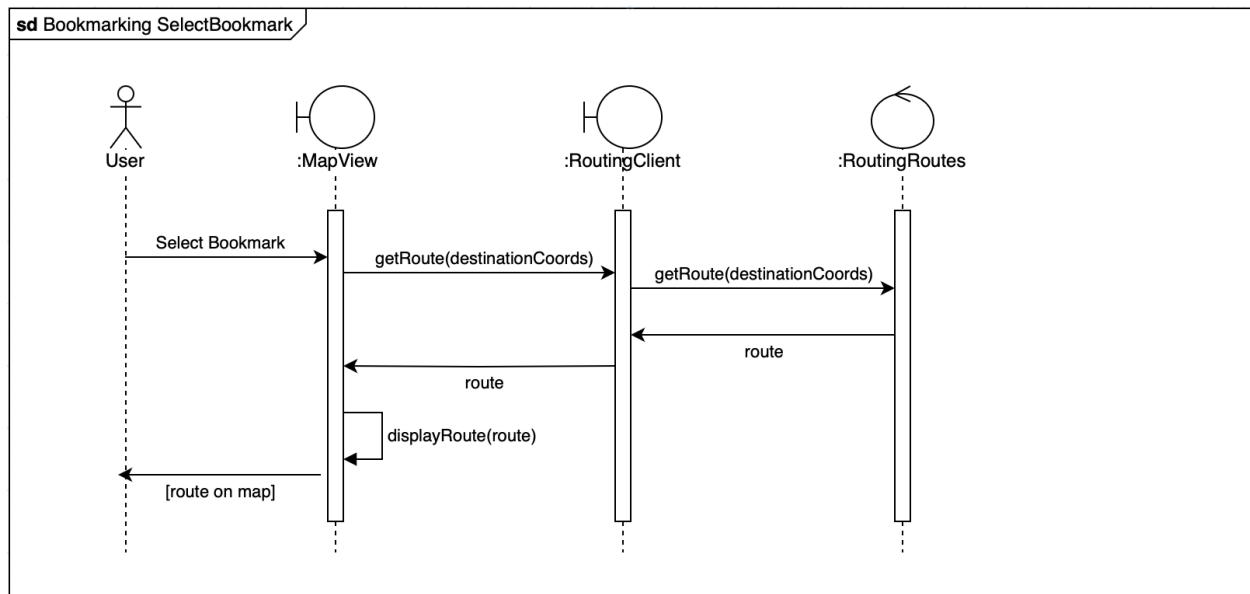


Figure 38: Sequence Diagram: Bookmarking (select)

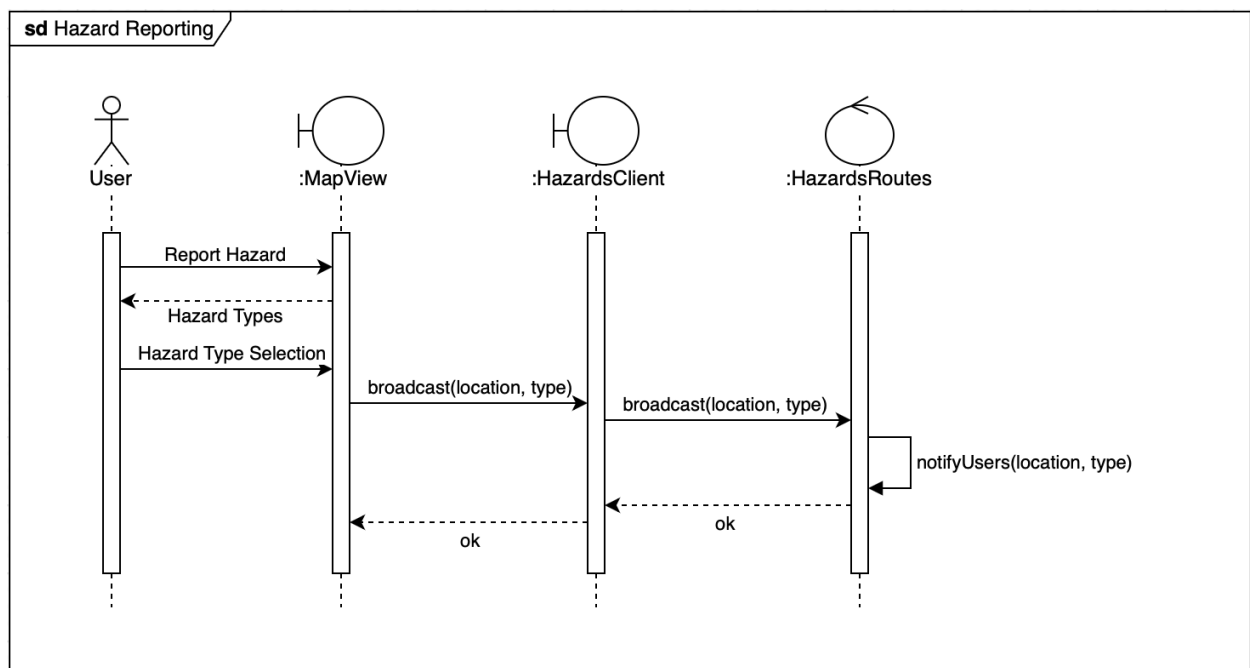


Figure 39: Sequence Diagram: Hazards (report)

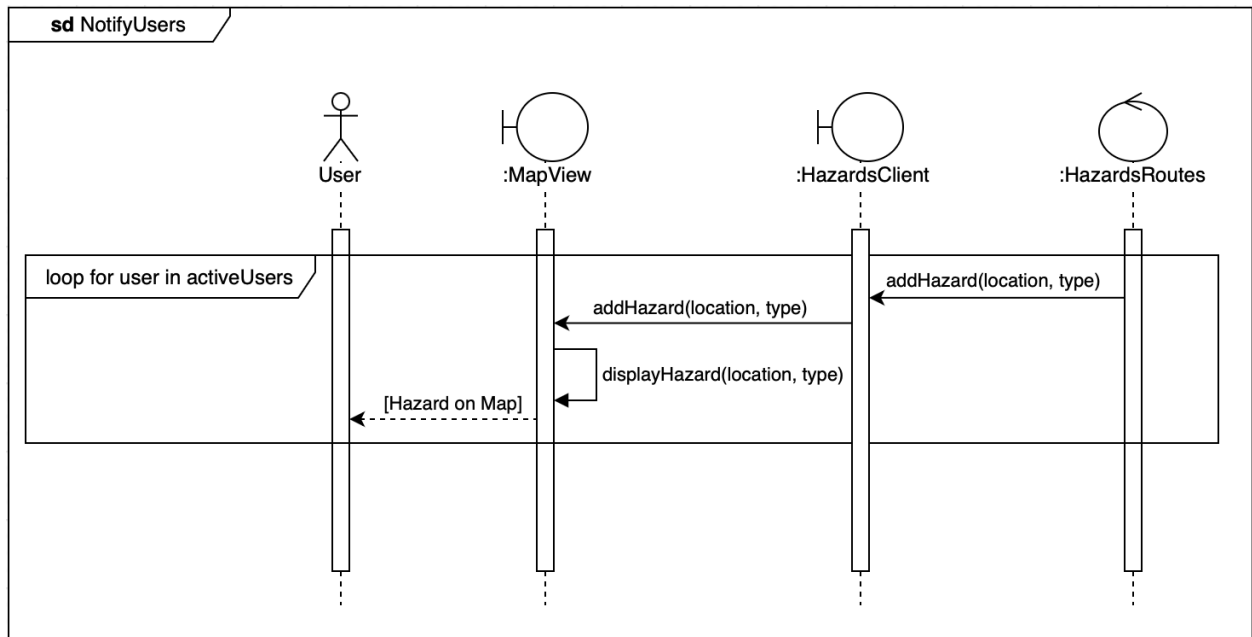


Figure 40: Sequence Diagram: Hazards (notify)

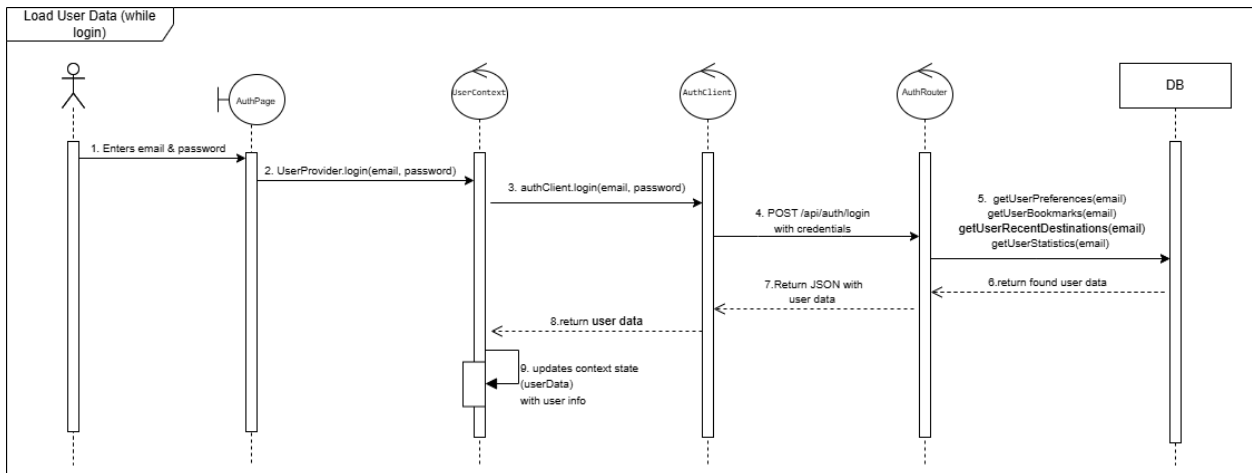


Figure 41: Sequence Diagram: User Data

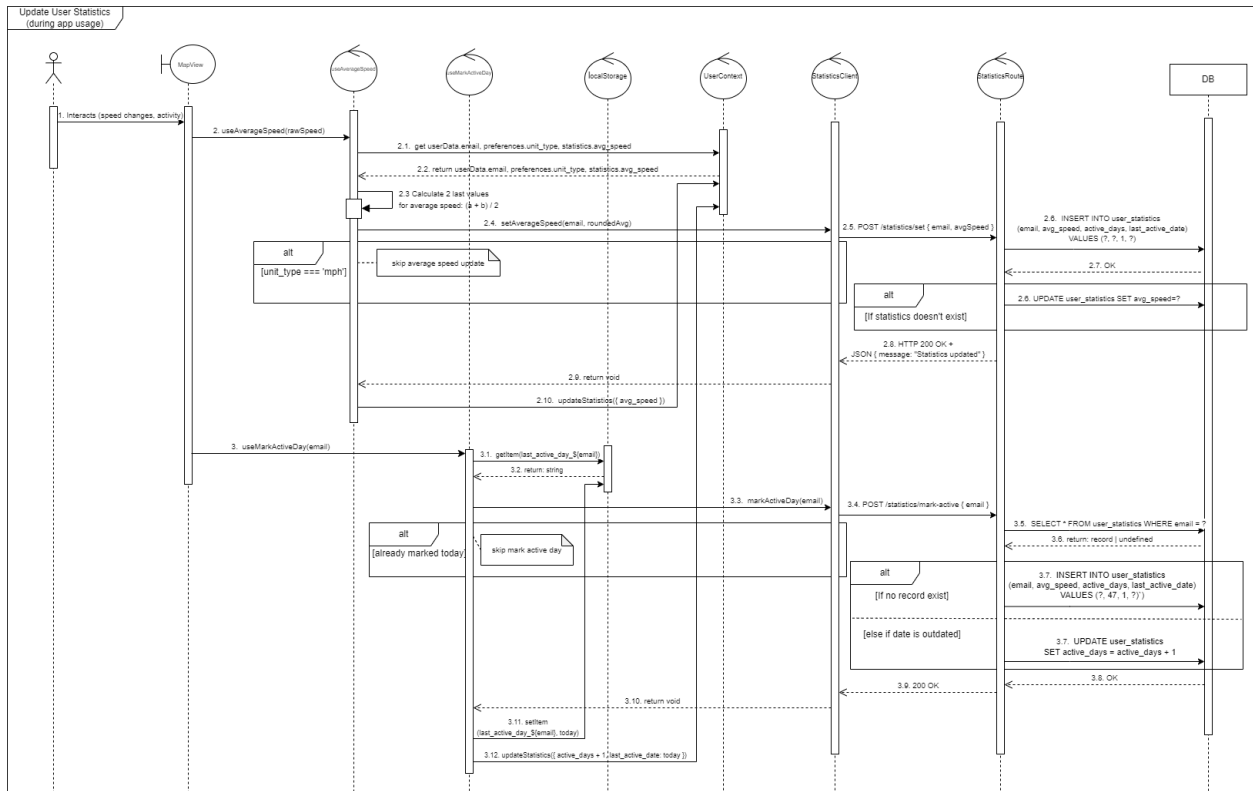


Figure 42: Sequence Diagram: Update Statistics

Development View

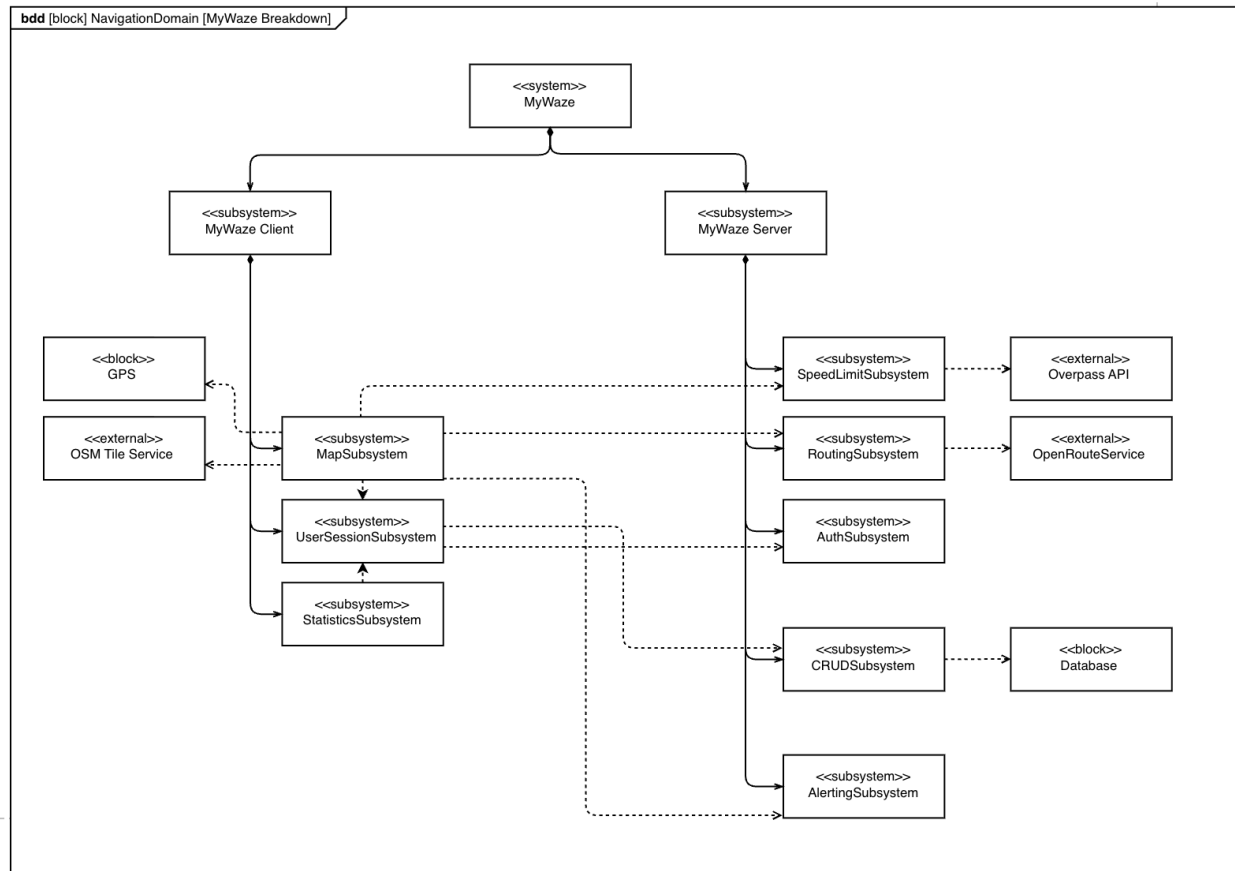


Figure 43: MyWaze Subsystems

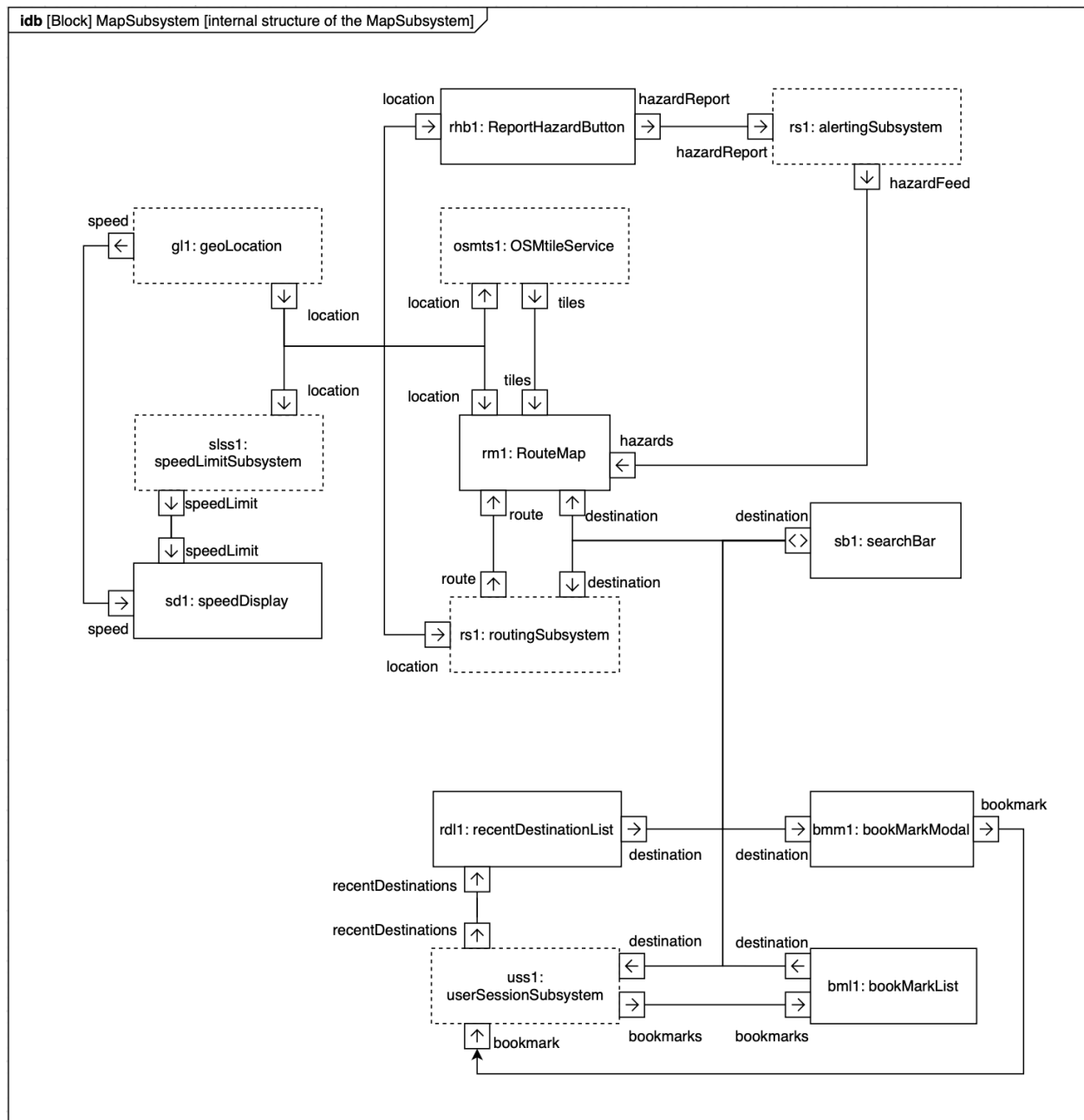


Figure 44: Internal Block Diagram: Map Subsystem

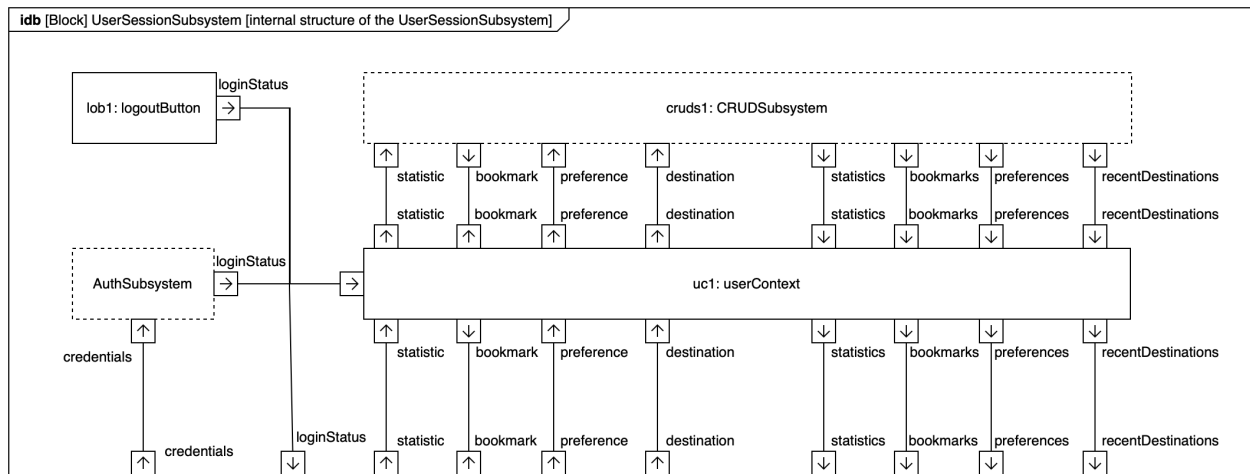


Figure 45: Internal Block Diagram: UserSession Subsystem

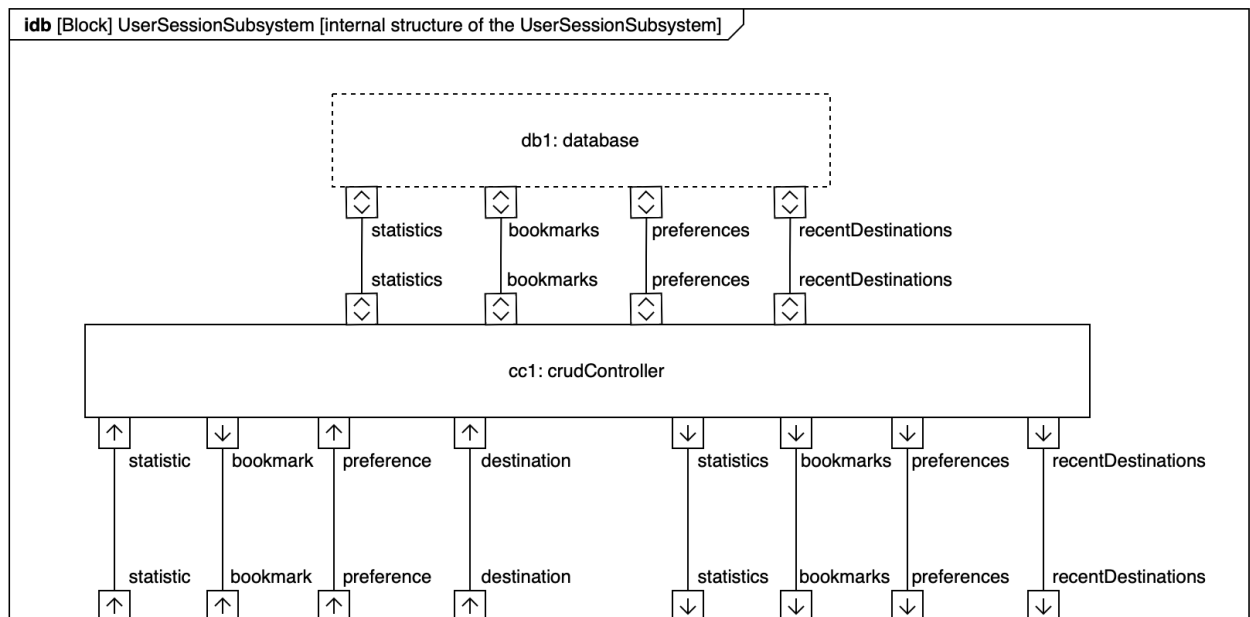


Figure 46: Internal Block Diagram: CRUD Subsystem

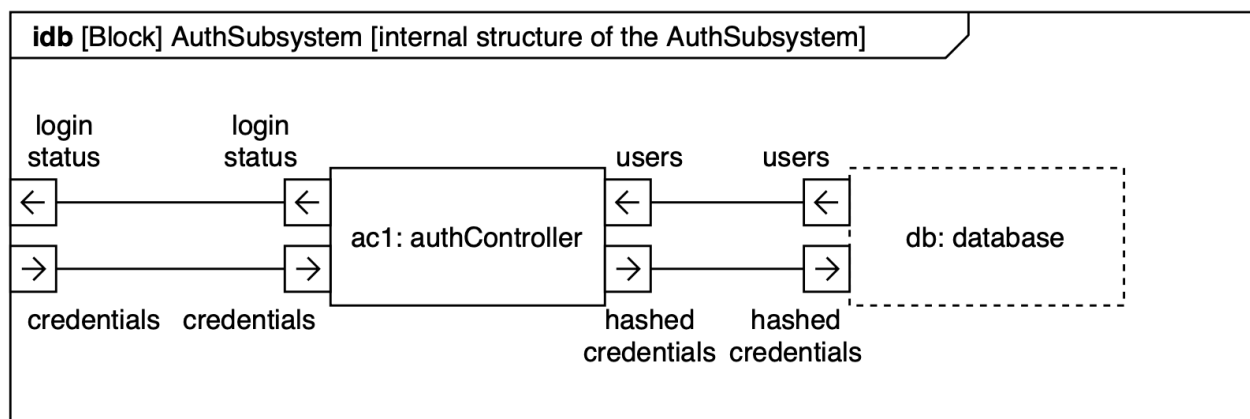


Figure 47: Internal Block Diagram: Auth Subsystem

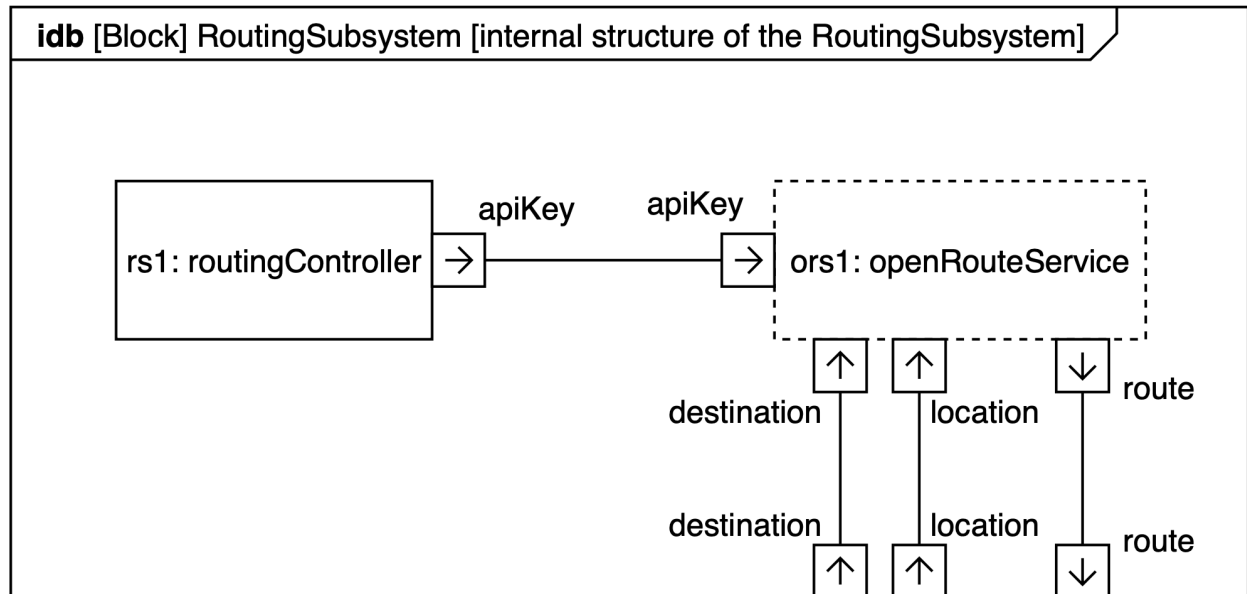


Figure 48: Internal Block Diagram: Routing Subsystem

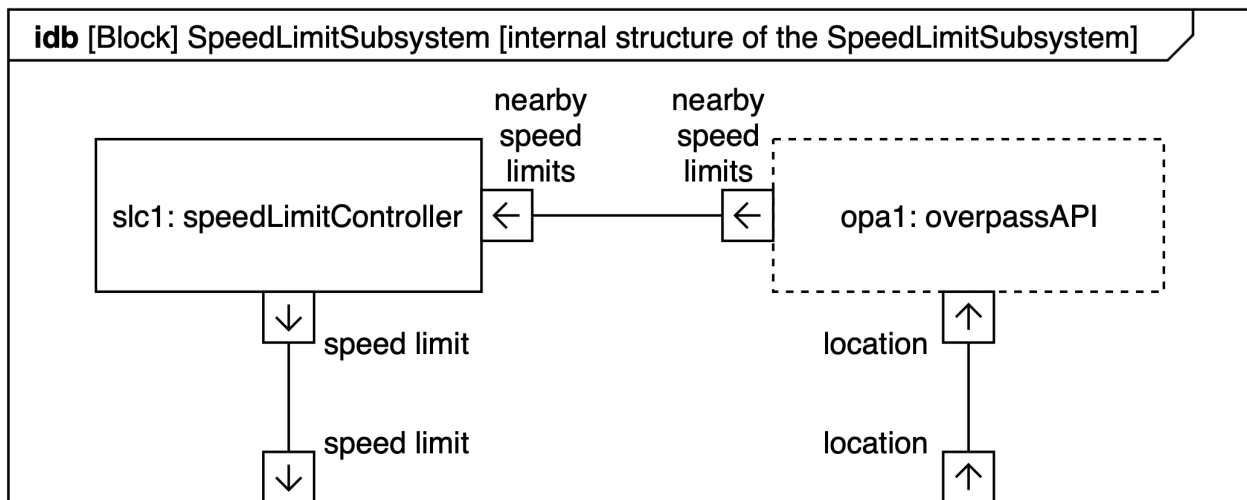


Figure 49: Internal Block Diagram: Speed Limit Subsystem

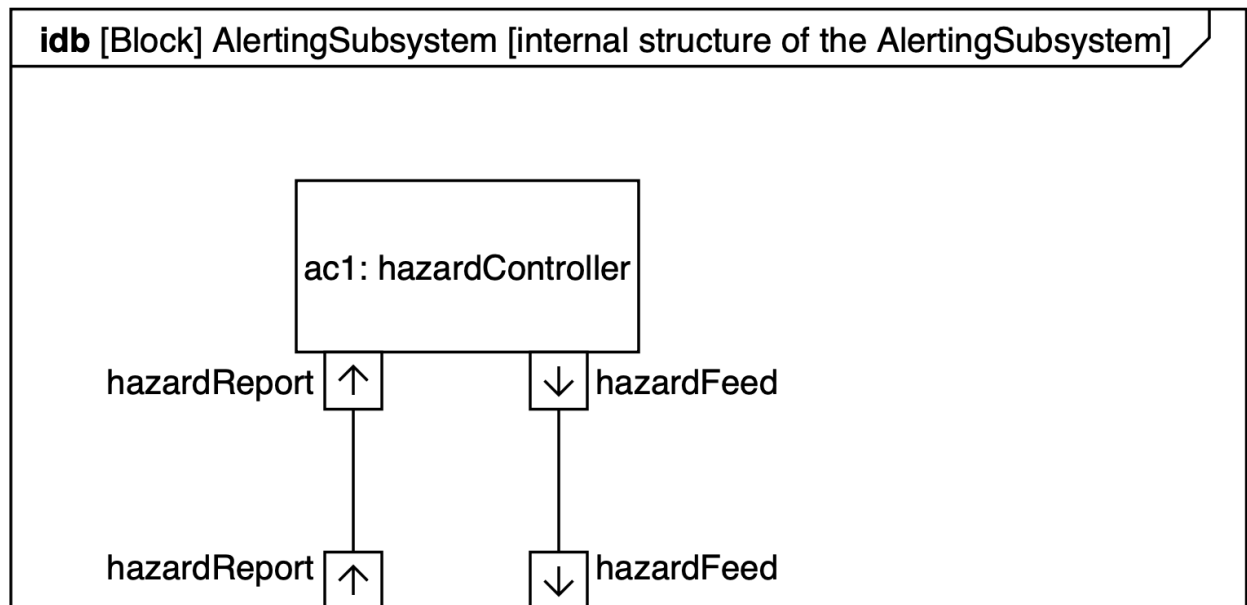


Figure 50: Internal Block Diagram: Alerting Subsystem

Variability Model

Feature Model

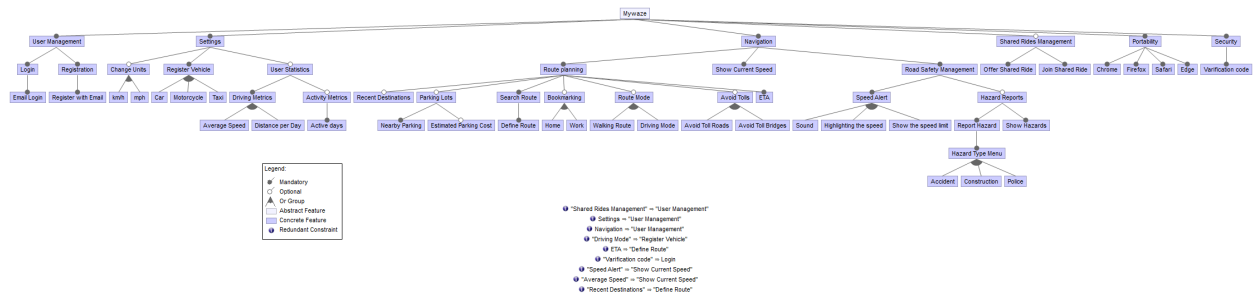


Figure 51: Feature Model

Configurations

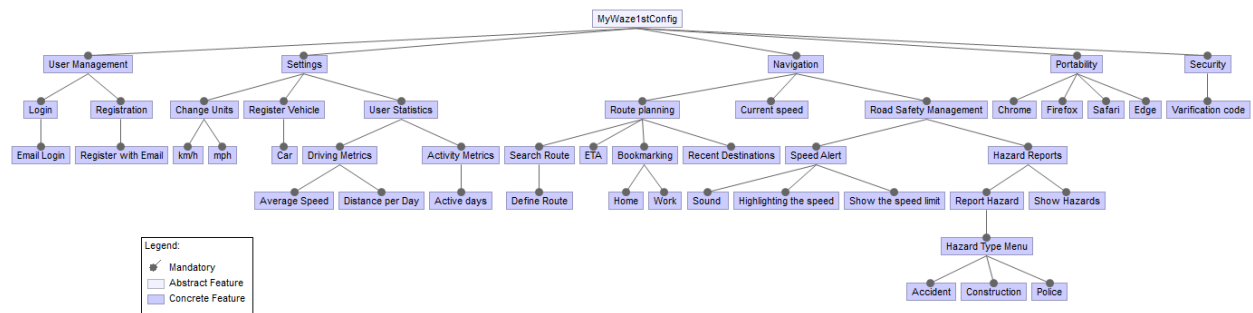


Figure 52: Configuration 1

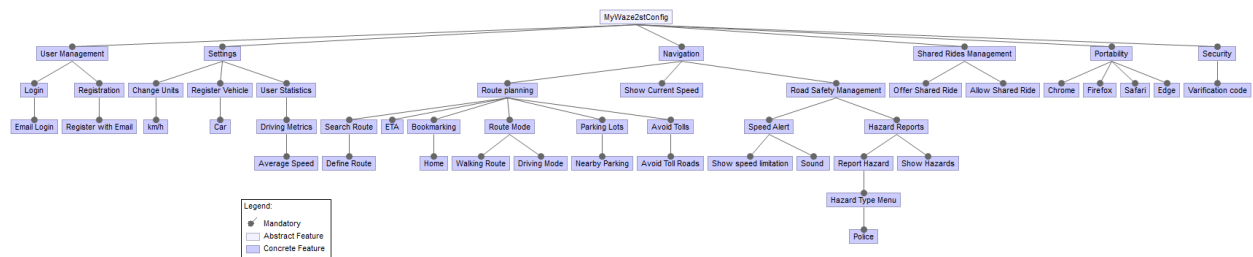


Figure 53: Configuration 2

Prototype

Running Instructions

Backend:

```
cd msp-mywaze-backend
npm i
npm run dev
```

Frontend:

```
cd msp-mywaze-frontend
npm i
npm run dev
```

The only prerequisites are a current version of **node** and **npm**. The database is implemented as SQLite, so no manual setup is necessary.

Routing uses OpenRouteService, which is quota-limited (daily) in the free version. If it stops working, please wait one day or change the variable `ORS_API_KEY` in `msp-mywaze-backend/src/routes/routing.ts` to your own (free) API key obtained from <https://openrouteservice.org>.

Features Implemented

Screen recordings of all features can be found in the `report/images/prototype` directory, or in the `readme` file in the top level of the repository structure.

Basic Features

- **Registration:** User can register and login on the home page.
- **Vehicle Type Registration:** User can configure their vehicle type in the drop-down user menu on the top right of the screen.
- **Define Route:** User can define a route by entering the destination name or address into the text field at the top center and clicking *Go*.
- **ETA:** ETA is displayed in a popup above the destination after a route is defined.
- **Speed Warning:** Current speed will turn red if the user moves over the speed limit. To test speed warnings on desktop, click on the speed value to generate a random speed around 50 km/h.

Advanced Existing Features

- **Bookmarking:** User can bookmark frequently used locations.
- **Hazard Reporting:** Users can report hazards encountered along the route. Reports are synchronized and shown across user sessions.
- **Recents:** Recently entered locations are shown for quick access.
- **Units:** Users can switch between metric and imperial units.

Advanced New Features

- **Stats:** Displays statistics for average speed, distance, and active days.